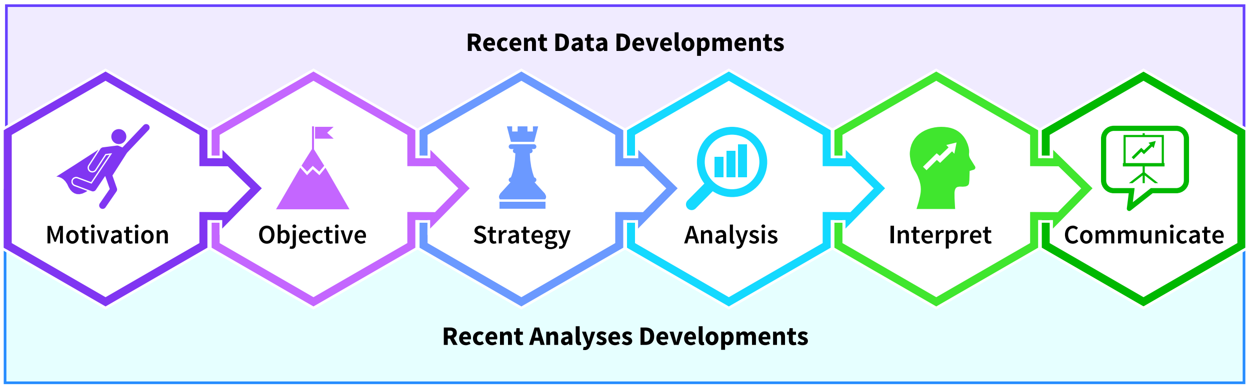
CHAPTER **10**

Recent Data and Analyses Developments in Accounting

Chapter Preview

This chapter reviews a variety of new data and analysis technologies that can add value and create opportunities for your professional work. These opportunities are limited only by your interest, creativity, and initiative. In fact, new data sources and analysis technologies are improving every step of the data analysis process.



Professional Insight: What Opportunities do Data and Analyses Developments Offer?

Neo recently finished his internship in a financial institution and summarized his introduction to the endless possibilities and opportunities available by pursuing a career in accounting.

> I discovered a passion for combining accounting and finance data during my internship. By analyzing accounting information trends, I could help define my client's capital needs and find the lowest cost capital sources. **I used diverse data sources and complex analyses to describe and predict my clients' financial value, cash flows, and capital needs to help me prescribe the best capital source alternatives.**
>
> I loved developing models to predict my client's cash needs and prescribe the sources and costs associated with their capital sourcing options. For example, I helped one client decide whether to extend ownership or borrow funds with prediction models. Analyzing their cash flows revealed they could afford to leverage more debt. **I created my models from data from the accounting system, financial statements, electronic contracts, such as leases and loans, and company-related social media posts, using both structured and unstructured data.**

These experiences fostered the creative skills necessary to build models that measure future performance and economic valuation, whether the client is a public, private, nonprofit, or government organization. **I feel that my work had purpose because I was helping clients find opportunities to increase their sustainable competitive advantages. That sense of purpose has reinforced my choice to study accounting.**

Chapter Roadmap

LEARNING OBJECTIVES	TOPICS	APPLY IT
LO 10.1 Describe data trends impacting the accounting profession.	• Data Volume • Data Variety • Data Velocity • Data Veracity • Data Value • Data Ethics	**Identify Data Characteristics** (Example: Managerial Accounting)
LO 10.2 Explain how technology developments are impacting data analysis in accounting.	• Value Creation • Data Mining and Smart Contracts • Process Automation and Process Mining • Continuous Auditing • Textual Analysis • Cognitive Technologies	**Use Process Automation to Analyze Financial Statements** (Example: Financial Accounting)
LO 10.3 Demonstrate how data and technology developments are adding value to professional practice.	• Accounting Information Systems • Auditing • Financial Accounting • Managerial Accounting • Tax Accounting	**Match Professional Practice Areas to Analyses and Technologies**

Data The Data tag appears in the chapter when the data for an example, illustration, or application are available on Wiley's online learning platform.

Data analytics software is continuously changing, and there may be more recent versions of the software referenced in this chapter. For more information access the accompanying video on Wiley's online learning platform.

10.1 Which Data Trends are Impacting Accounting Practice?

LEARNING OBJECTIVE 1

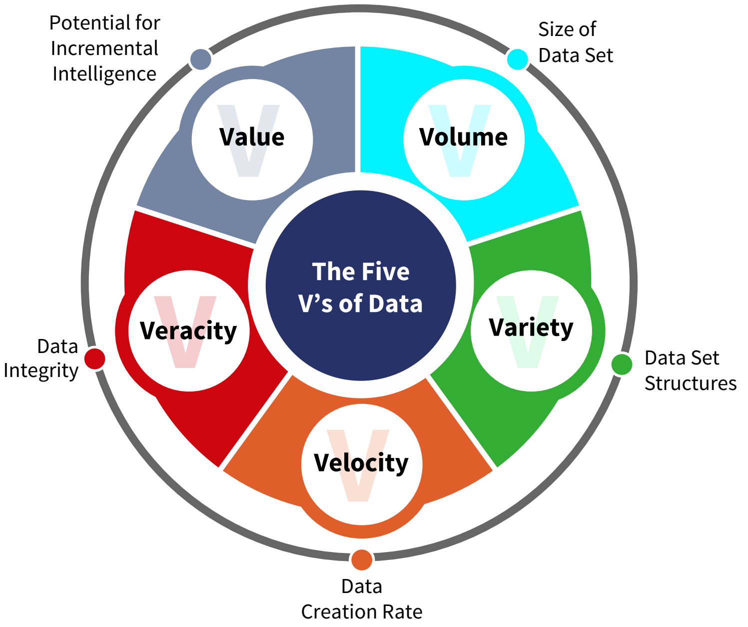
Describe data trends impacting the accounting profession.

More data from new and diverse sources are available every day. Consider the online course management platforms used by college students around the world. These learning management systems can track each selection a student makes during online learning. If your objective is to do well in a course, then this data can be analyzed to provide valuable feedback about which concepts you still need to master so you can focus on learning them.

Accountants are also encountering new data options and characteristics from sources both internal and external to their organizations. Whether that data is valuable depends on their objectives. If the objective is improving tax planning, then internet chatter data about upcoming tax policy changes could be valuable. Of course, this same chatter may not be valuable if the objective is to evaluate shipping costs for the year.

As accountants perform analyses using more data and increasingly powerful computers and analysis tools, they must consider how increases in data volume, variety, velocity, and veracity add value to their analyses. **Illustration 10.1** defines these five data characteristics and provides examples in each professional practice area.

ILLUSTRATION 10.1 The Five Data "Vs"



Characteristic	Examples
Volume	In AIS: • Process flow logs • Security alerts • System error logs • Transactions • Firewall traffic • System and application access logs
Variety	In auditing: • Codes, approval stamps, text, pictures, dates, and recorded values • Fair market values at year end
Velocity	In financial accounting: • Cash transactions • Stock price changes • Social media content
Veracity	In managerial accounting: • Transactions • Prices and costs • Vendor and customer data • Sensor data from smart machines
Value	In tax accounting: • Local, national, and international tax changes • Sales tax jurisdictions and rates

This section covers the impact of each data characteristic on accounting data analytics, then explains the ethical implications accountants consider throughout the data analysis process stages.

Data Volume

Imagine if the number of clothes in your closet grew exponentially each year. Depending on your closet's size, you might run out of storage space and struggle to locate the items you need when you need them. Adding storage space and proper organization could cost you both time and resources.

Accountants are experiencing similar increases in data, which is leading to more data storage and organization costs and more data strategy options. **Data volume** refers to the amount of data selected for an analysis project. Data volume is typically limited by the computer and analysis tool capacities required for the data analyses.

Increases in Data Volume

Business operating cycles in large organizations are currently capturing and storing millions of sales, purchases, receipts, and payments each day, in addition to email, social media, and internet traffic content. Organizations are also storing more data than ever before. In fact, experts believe the volume of data produced by organizations will continue to increase exponentially in the next decade (**Illustration 10.2**).

ILLUSTRATION 10.2 Organizations are Producing More Data

Two factors are contributing to this:

1. Technologies that are connecting and integrating data from traditionally separate systems, such as the AIS, internet content, and smart device sensors.
2. Decreasing computer costs and increasing computer memory and processing capabilities make it easier for organizations to capture, store, and analyze more data.

Of course, with more data comes more costs. Businesses need more computer hardware and software capacity, data privacy, security expertise, and internal controls. To address these challenges, organizations are turning to new solutions for storing and processing their large data sets.

Developments in Data Storage

A popular solution for storing high data volume are cloud data services. **Cloud data services** provide secure data storage on large servers, using the organization's own resources or resources provided by third parties. You might be familiar with some of the data clouds used by college students, such as Google Drive, Microsoft's OneDrive, pCloud, sync.com, mega.io, and box.com. What are the benefits of cloud data services? Benefits include:

- Easily expandable data storage capacities.
- Expert data management.
- Secure data, hardware, and software.
- IT expertise and advice.

There are three types of cloud data services. The key difference between them is if and how the cloud service provider's servers are shared:

- **Private clouds** are restricted access data storage centers created for the data of one organization or shared between an agreed upon set of companies. Private shared clouds often include a single large company, a group of supply chain partners, or non-competing independent companies from different industries. Private clouds, provided by third parties such as HP Enterprise, VMWare, Microsoft, Cisco, and Amazon, are the fastest growing cloud sector.

- **Public clouds** securely store data from multiple companies on shared servers using virtual server data separators. Leading public cloud service vendors include Google Cloud Platform, Amazon Web Services (AWS), and Microsoft Azure.
- **Hybrid clouds** offer both private and public cloud data storage, serving the needs of the security and use characteristics of the data involved. Amazon and Microsoft are two leading vendors providing hybrid cloud solutions.

APPLYING CRITICAL THINKING 10.1: Evaluate Cloud Data Services

Use critical thinking when deciding which data cloud type best serves your data needs by researching prices and services offered by viable data cloud vendors **(Knowledge)**.

The best data cloud can also be selected by considering the following factors **(Alternatives)**:

- The volume of data you need to store.
- How often and how much of the data needs to be added, modified, and retrieved.
- Data privacy and data security requirements.
- Your budget for cloud data services.
- The reputation of the vendor.

While organizations might prefer to store their data in private clouds, both private and public shared cloud services can provide significant cost savings. Think of it as living in a building with several separate apartments. Some people might prefer to live in a house by themselves, but a shared structure often means saving money on housing costs. **Illustration 10.3** summarizes the typical benefits and costs of each type of data cloud.

ILLUSTRATION 10.3 Comparing Cloud Data Services

Data Cloud	Benefits	Costs
Private Clouds		
Data storage for a single organization.	• Offers high data privacy and data security. • Offers high data interactivity. • More customization and adaptability are possible.	• Can be the most expensive option. • May be difficult to manage if developed in-house without hiring staff with cloud expertise.
Data storage that is shared by a set group of non-competing organizations.	• All the benefits of single organization storage, especially for data interactivity between supply chain partners. • Sharing data storage makes it a less expensive option for each company.	• Can have some restrictions on customization and adaptability. • May involve shared contracts and joint revisions.
Public Clouds		
Public clouds are shared by many organizations who can easily come and go from the public cloud.	• Lowest individual company cost. • Fastest implementation. • Most appropriate for small businesses.	Least adaptable or customizable cloud data service.
Hybrid Clouds		
Include both private and public cloud virtual server sections.	The benefits and costs associated with both private and public clouds also apply to hybrid clouds. Hybrid clouds allow organizations to choose which data subset is best stored in a private cloud, while storing the rest in a public cloud for the cost efficiencies.	

Some vendors of cloud data services also offer cloud computing services, providing the software computing power that is often needed when organizations process and analyze large volumes of data. A recent market strategy pivot is the movement of high tech and internet service companies towards also providing private and public cloud data and computing services, leveraging their existing data storage, processing, and security service reputations.

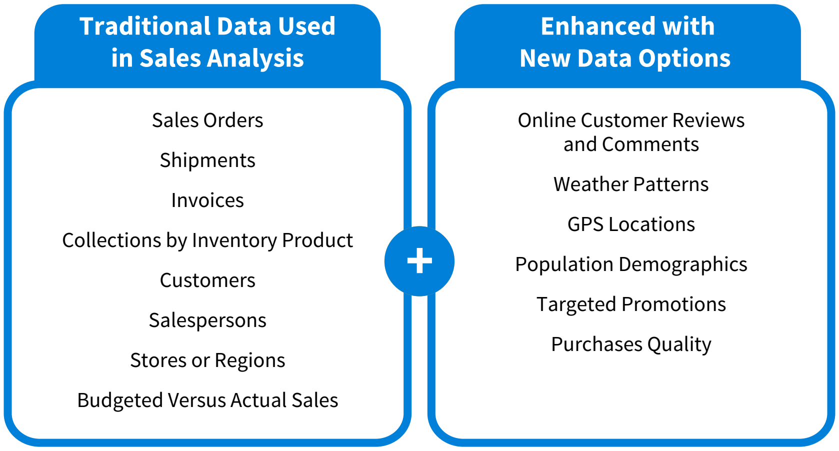
Data Variety

Data variety refers to the diversity of data structures and measurement scales in the data we want to analyze. Let's return to the wardrobe example. We can describe our clothes with a variety of structured data fields: categorical scale fields such as color and type (shirts, shoes, etc.), ordinal scale fields such as size, interval scale fields such as how long we have owned the garment, or ratio scale fields such as how much it originally cost.

Accountants have traditionally selected and analyzed structured data that are easily represented in rows and columns, such as categorical account codes, interval transaction dates, and ratio values for the transaction amounts and account balances. Recently, accountants are combining structured data with unstructured data, including text, smart device sensor readings, and pictures and graphics, to increase the value created from their analyses.

Illustration 10.4 provides examples of unstructured data that can be combined with traditional transaction data from customer orders, inventory, shipments, invoices, and collections.

ILLUSTRATION 10.4 Analyzing the Sales Function: The Impact of Data Variety



Valuable business intelligence can be created by combining unstructured data with traditional structured data for analysis:

- Shipping truck GPS data can be integrated with AIS data to find the most efficient delivery routes, reducing truck wear and tear by reducing mileage driven, reducing fuel costs, and improving delivery time.
- Integrating weather data with sales transaction data to anticipate customer traffic patterns and allow for better service.
- Combining customer reviews with inventory turnover to assess vendor and product effectiveness.
- Integrating GPA mapping data with cell phone location data can evaluate or develop targeted advertising campaigns or for marketing intelligence about future products.

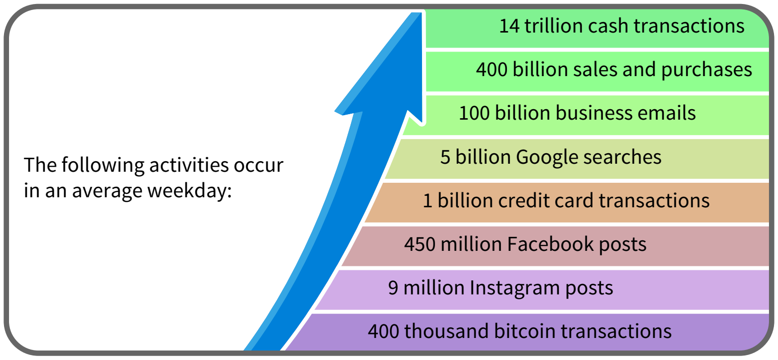
APPLYING CRITICAL THINKING 10.2: Consider the Impact of Data Variety on Data Strategy

As the variety of available data and how they are combined increase, the success of your data and analysis strategies can be improved by selecting an analysis that is appropriate for your data's measurement scales. One option is transforming unstructured data into structured data scales to increase your analysis options. For example, categorizing text comments as "positive" or "negative" transforms unstructured text fields into structured categorical data which better addresses the needs of marketing and sales managers. These two categories can be tested for correlation with other data, such as delays in shipments, to better understand how to best serve the stakeholder providing the comment. (**Stakeholders**, **Purpose**)

Data Velocity

Most of us have experienced how rapidly information moves through social media platforms such as Instagram and Twitter. **Data velocity** refers to the speed with which new data points are generated. The higher the data velocity, the faster data volume grows (**Illustration 10.5**).

ILLUSTRATION 10.5 Data Velocity Continues to Increase



Accountants now regularly use high velocity data. Internal high velocity data can come from various sources:

- Equipment like HVAC (heating, venting, and air conditioning) and smart sensing and controlling devices for cost and quality control purposes.
- Business process tags indicating throughput progress for operational control.
- Tags on confidential data and sensitive word use for communications originating inside the organization to manage legal and market risk.

High velocity data can also come from external sources:

- Mobile device location alerts indicating when customers are nearby for targeted advertising.
- Location, traffic, weather, and construction alerts for employees operating delivery or service trucks to optimize routes and service.
- Tags on sensitive words in tweets, posts, texts, emails, and blogs that originate outside the organization to identify legal risks and market intelligence.
- Weather and geographic map data for sales forecasting and transport safety.

Data velocity is also related to the terms big *data* and *internet of things*. Coined in 1987 to help market new computing systems, big data refers to how high data velocity can quickly increase data volume. When traditional AIS data is integrated with high velocity data from smart devices, the resulting high variety data set is sometimes called the internet of things. Many technological advances, including the cloud data services discussed earlier, have been developed to provide solutions for the challenges experienced in storing and processing high volume, high variety, and high velocity data.

APPLYING CRITICAL THINKING 10.3: Tips for Using High Velocity Data

While there are many tweets on Twitter every day, frequently only some are relevant to the purpose of an analysis:

- A common tool for evaluating high velocity data as they are generated is using an "if, then" condition and only storing the data that pass the specific condition. For example, if a tweet mentions your company's name, then that tweet will be stored **(Purpose)**.
- The value of high velocity data can be short-lived. The same "if, then" filters that identify which data have value can also be used for data purging when that value expires. Being alerted that there may be hail in the next hour can protect assets from hail damage, but hail alerts from yesterday will not protect assets today. (However, this data could still be valuable to insurance companies who are estimating future monthly hail damage claims) **(Risks)**.

Data Veracity

We often rely on data to help improve our decisions. Sometimes the data needs to be accurate for our objective, and other times data just needs to provide general direction. A weather forecast does not have to be exactly accurate to help us dress appropriately. Alternatively, an assignment due date or an invoice due date must be precise to ensure we complete the assigned work or pay the invoice on time.

Accountants need reliable data from trusted sources to make professional judgments and decisions. **Data veracity** refers to the reliability, or the integrity of the data we select for analysis. Data veracity depends upon three things:

1. Whether the data will likely provide insights for the analysis objective, either by itself or as a component in an information model, such as a net income calculation.
2. How accurate the data must be to generate useful insights for the objective.
3. How complete the data must be to generate useful insights for the objective.

Data veracity is often considered when ranking data and analysis strategy alternatives. Assume the objective is to learn whether prior customers become repeat customers or if they purchase from competitors instead. One data alternative might be analyzing the content of customers' related social media postings. Another choice is analyzing the last two years of sales data:

- Social media postings, when authentic, describe current or past customer experience satisfaction (or lack thereof), and may reveal if they are purchasing from competitors. The postings do not directly measure whether they would choose to be repeat customers in the future.
- Past sales data probably have higher data integrity and better identify prior repeat customers. We do not know if prior choices predict future choices or if customers are purchasing from competitors.

APPLYING CRITICAL THINKING 10.4: Evaluate Data Veracity

Accurately evaluating data veracity helps you select the best data and analysis strategies and better interpret the results. Your biases, assumptions, and self-interests, or those of others who selected or analyzed the data, may impact how you judge the data's veracity: **(Risks)**:

- Data veracity can be over-rated by the person selecting, preparing, and analyzing the data, as most people defend their own choices.
- Data veracity can also be under-rated when the data or results conflict with what is believed or expected about the phenomena, and when you do not know what, if any, controls were used when the data was selected, prepared, or analyzed. For example, if you expected sales revenues to increase, you may discount data which challenges your expectation.

Data Value

Data value refers to the benefits the data provides given the analysis' objective. Our clothes have a different value depending on the day's agenda. Professional attire is worth more when we have an interview, go into the office, or visit clients. Lounge wear, on the other hand, is more valuable when working remotely from home or enjoying personal time. Accountants' judgments regarding data value depend on the project's objective:

- If the objective is to describe profits or evaluate profit trends, then revenue and expense data are valuable data.
- When diagnosing the causes of decreasing operating cash flows, then data from the current asset and current liability accounts are valuable.
- If the objective is to predict bankruptcy risk, then liquidity and solvency data, such as a company's cash flows and all liabilities, would have data value.
- When prescribing asset utilization strategies, market data informing about future revenues and expenses have data value.

Let's illustrate data value with a managerial accounting example. Assume you work for a company that manufactures and sells a variety of computer keyboard product lines. You are considering dropping a product line because it is operating at a negative net income. Data that allows separating traceable fixed expenses from common allocated fixed expenses would make it possible to calculate segment contribution margin in addition to the product line net income (**Illustration 10.6**). (**Data** See How To 10.1 at the end of the chapter to learn how to create a comparative income statement in Excel, and then learn how to use an automated process to create it in How To 10.2).

ILLUSTRATION 10.6 Comparing Data Value of Net Income and Segment Margin

	Income Statement			
	Product A	Product B	Both Product Lines	Only Product A
Sales Revenues	$720,000	$1,200,000	$1,920,000	720,000
less Variable Costs	432,000	960,000	1,392,000	432,000
= Contribution Margin	288,000	240,000	528,000	288,000
− Avoidable Fixed Expenses	210,000	225,000	435,000	210,000
= Segment Margin	78,000	15,000	93,000	78,000
− Unavoidable Common Fixed Expenses	42,000	42,000	84,000	84,000
= Net Income	$ 36,000	$ (27,000)	$ 9,000	$ (6,000)

Segment margin provides incremental intelligence value over the net income measure. This is because if the segment margin result is positive, then that product line of computer keyboards is generating wealth to cover the common fixed expenses. The final column in Illustration 10.6 shows that if Product B is dropped, the company would lose the $ 15,000 of segment margin generated by Product B and would make a loss ($ 6,000). This example shows that the separation of traceable and common fixed expense data has high data veracity.

Economists often measure the value of data by whether it is likely to confirm or change expectations about an objective's phenomena, (e.g., the net income or the segment margin) or by how it reduces uncertainty about the phenomena (e.g., the dispersion of the data points). In fact, data value is often not fully understood until analysis hypotheses are tested and the results of the analysis are available.

Data Ethics

Using personally identifiable data can raise ethical issues for individuals and organizations. On a daily basis, accountants interact with the personal and confidential data of clients, employees, customers, investors, and creditors. These stakeholders rely on accountants' professional ethics to make judgments and communicate information while protecting their privacy and maintaining the confidentiality of their data. Unintentional and intentional personal data violations can occur in various ways:

- Collecting and sharing more than the minimally needed private data without explicit consent.
- Failing to provide easy and clear opt-in and opt-out consent for data collection, sharing, and deletion (pre-checked boxes on websites do not provide explicit consent).
- Limiting access or opportunities for those who opt-out of sharing their personal data.
- Inadequate data access controls, such as encryption, which lead to access violations.

Many specific types of data, such as personally identifiable information, client financial records, employee pay and health records, and a company's future strategic plans, are protected by laws and regulations.

Privacy Laws

Illustration 10.7 summarizes the leading data privacy laws in the United States.

ILLUSTRATION 10.7 U.S. Data Privacy Laws

Law	Description
The Privacy Act of 1974	Restricts use of data held by government agencies.
Health Information Privacy Protection Act of 2013 and the Health Insurance Portability and Accountability Act of 1996 (HIPAA)	Restricts the sharing of individual's health care data, with fines up to $10,000 per violation.
The Fair Credit Reporting Act (FCRA)	Limits and protects the credit-related personal data collected and shared.
The Family Educational Rights and Privacy Act (FERPA)	Protects the privacy of educational records.
The Gramm-Leach Bliley Act (GLBA)	Requires loan and investment service providers to disclose personal data sharing.
The Electronic Communications Privacy Act (ECPA) and Patriot Acts	Restricts when and how individuals' private data communications can be legally surveilled.
Children's Online Privacy Protection Act (COPPA)	Restricts personal data collected from minors and extends to third parties' data use.

U.S. states are also passing their own data privacy laws at varying speeds:

- California, Illinois, and Colorado were the first states to pass comprehensive consumer data privacy laws.
- Other states, such as Massachusetts, New York, Hawaii, Maryland, and North Dakota, are moving forward with both partial and comprehensive data privacy law proposals that are currently in committee at their state legislatures.

The progress of these laws can be tracked on websites such as the International Association of Privacy Professionals. To help accountants navigate this data privacy landscape, there are ethical standards, privacy policies, and codes of conduct.

Ethical Standards and Privacy Policies

Ethical standards for using personal data assume that accountants will stay current with the relevant data privacy laws and regulations. Inappropriately sharing a client's confidential contact information, financial status, or tax liability can result in losing client trust and a damaged professional reputation. Some data privacy infractions may have much more serious consequences, such as losing professional licenses or other sanctions:

- The Institute of Management Accountants' (IMA) ethical standards require managerial accountants to act with competence, confidentiality, integrity, and credibility when interacting with their organization's data, particularly the personally identifiable data of employees and customers.
- The Association of International Certified Professional Accountants (AICPA) has similar due care ethical standards for CPAs using private and confidential data. CPAs also have ethics requirements about their objectivity and avoiding conflicts of interest.
- Tax accountants, whether certified or not, must abide by the strict data confidentiality requirements of the Gramm-Leach-Bliley Act. The IRS code sections 6013 and 7213 require tax accountants and IRS employees to maintain an individual's tax return data privacy and security. Violations risk misdemeanor charges, fines up to $5,000, and imprisonment up to five years per infraction.

APPLYING CRITICAL THINKING 10.5: Learn About Data Laws and Regulations

Accountants must know the cultural, legal, regulatory, and transparency expectations for the collection, storage, and uses of personal data, both in the United States and globally. For example, if your company is expanding to the European Union, then per the GDP you must offer those customers the ability to opt out of having their private information stored after a sales transaction is fulfilled. You can find some of the current laws and regulation at the following sources **(Knowledge)**:

- The European Union's General Data Protection Regulation (GDPR)
- Center for Internet Security's critical security controls
- NIST Critical Infrastructure Security Framework
- National Conference of State Legislatures' Security Breach Notification Laws

Ethics and the Effects of Data Velocity and Volume

Whenever you activate your navigation app's location feature, either in your vehicle or cell phone, any data partner of that navigation app can access your location and movements throughout the day. This example of high data velocity and volume highlights new complexities when developing and maintaining compliant data privacy and security systems.

Data privacy is especially an issue for accountants when using internet browsers to access clients' or organizations' private data. A single website visit can place tracking cookies, as well as many third-party cookies, on our computers. If not immediately removed, these cookies can capture data retrieved from future website pages, username and password keystrokes, and cursor movements.

In the United States it is legal for apps or cookies running in a browser's background to collect, use, and share data without explicit permission if the information collected does not include individual user identifiers and is not used for identity theft. However, the risk of personal data being gathered exists. An easy and cost-free preventative control for this is to go into the browser settings and clear stored cookies (also known as clearing the browsing history) at the beginning and end of every browser session. Of course, the most secure solution is accessing private data from within the security of a virtual private network.

Considering data volume, variety, veracity, velocity, value, and ethics within the context of your analyses' objectives can help you make the best choices when designing data and analysis strategies. Next, let's learn about the power we can harness from recent technological developments in analysis tools.

Apply It 10.1
Identify Data Characteristics

Managerial Accounting You an accountant for Limitless Outdoor, a leading outdoor gear company. The objective of your data analysis project is to learn more about your company's sales and product quality from last year. For each data field, indicate which data characteristics best describe it. (More than one data characteristic can be used to describe a data field.)

Data Field	Data Characteristic
Sleeping bag fabric durability rating from quality testing.	
Sales revenues from repeat customers and new customers during the last 12 months.	
Cell phone data captured from bicycle rides using a navigation app.	
Variability of manufacturing equipment performance every five minutes during use.	
Naturally occurring bundles of camping equipment and outdoor clothing sales each month.	
Hourly outside temperatures during weekend days when most customers are using the equipment.	
Customer comments about their outdoor product satisfaction on social media platforms.	

SOLUTION

Data Field	Data Characteristic
Sleeping bag fabric durability rating from quality testing.	Value
Sales revenues from repeat customers compared to new customers during the last 12 months.	Veracity and value
Cell phone data captured from bicycle rides using a navigation app.	Volume and variety
Variability of manufacturing equipment performance every five minutes during use.	Volume
Naturally occurring bundles of camping equipment and outdoor clothing sales each month.	Value
Hourly outside temperatures during weekend days when most customers are using the equipment.	Volume
Customer comments about their outdoor product satisfaction on social media platforms.	Variety and volume

(This is a suggested solution; other relevant insights might also be correct.)

10.2 Which Recent Analyses Developments are Accountants Adopting?

LEARNING OBJECTIVE 2

Explain how technology developments are impacting data analysis in accounting.

Accountants can efficiently analyze more data than ever before by using established analysis tools such as Excel, as well as recently developed tools powerful enough to process a variety of data and at higher volume. In addition to tremendous analysis power, these tools have advanced visualization features that are improving how accountants analyze, interpret, and communicate analysis results.

Accountants often perform data analysis to prepare and verify appropriate recording of transactions and reporting of business operations and financial position. They are now also performing more data analysis to generate valuable business advice and professional judgments for their internal and external stakeholders. Several leading technologies are making this possible.

Value Creation

Accountants have a unique competitive advantage for creating value for their stakeholders since their perspectives integrate knowledge across disciplines. Thoughtful data analyses can help improve resources and processes to increase operational competitiveness and achieve financial objectives (**Illustration 10.8**).

ILLUSTRATION 10.8 The Data Analysis Value Creation Chain

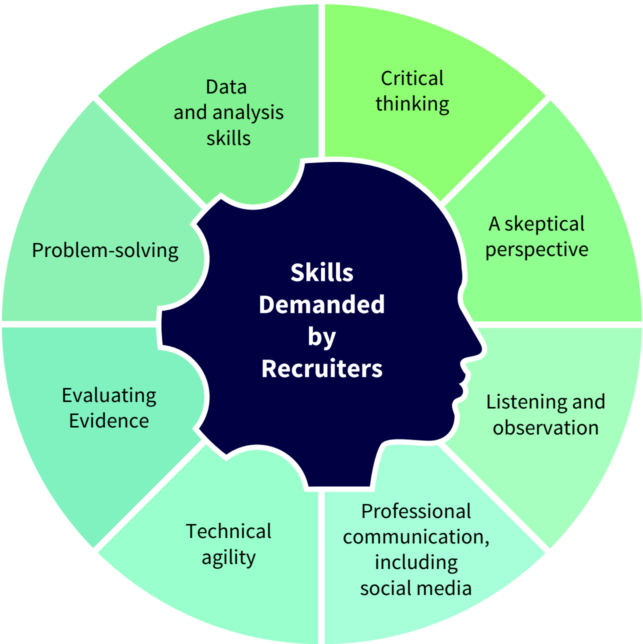
These opportunities have led to changes in how professional accounting organizations and businesses present themselves:

- The Association of International CPAs (AICPA) and the Institute of Management Accountants (IMA) have recently rebranded how their members serve their organizations and society. They have declared that their primary purpose is to create value for stakeholders, which include investors, creditors, customers, suppliers, regulators, and management and employees.
- Similarly, the large international accounting firm PwC recently moved away from the traditional compliance-facing labels of tax accountant and auditor. PwC's operations in the United States have been reorganized to combine the traditional audit and tax regulatory expertise services into "Trust Solutions." PwC has further rebranded their professionals as transparent, trusted business advisors focused on creating value through recommending sustainable growth outcomes for their clients and the world.

Accounting firms large and small are quickly changing their business models to present themselves as expert advisors who create value by performing data analyses to improve governance, operations, and compliance strategies for their clients. By following recommendations for better resource allocations and process improvements, organizations can improve access to capital, trading partners, and performance. For example, data analytics that reveal which product or service revenues are most associated with net income increases can grow both the client's and the public's trust in accountants' expertise and opinions.

Let's use a university example to illustrate how recent developments in data analysis have impacted value creation. Historically, faculty at institutions of higher education have evaluated internal student performance data to gain insights into how to improve their instruction. Access to new data options and analysis tools now allow colleges to analyze data from external stakeholders such as alumni, recruiters, advisory boards, and professional associations to learn which skills and knowledge are demanded by the profession (**Illustration 10.9**).

ILLUSTRATION 10.9 Accounting Knowledge and Skills Demanded by Recruiters and Professional Association Standards



Accounting programs can now analyze a variety of valuable data from alumni and recruiter surveys, interviews, and high volume and velocity data from social media platforms, such as LinkedIn and Facebook, to create courses that produce successful graduates. They can also evaluate an accounting program's effectiveness by external metrics, such as how many internships turned into full time post-graduation job offers and by measuring how long it takes alumni to be promoted after graduation.

Similarly, accounting professionals are using more types of data and newer analysis tools to create value:

- Data mining tools and electronic contracts for revenue recognition and lease accounting.
- Routine process automation and process mining tools for improvements in process efficiency and control.
- Continuous auditing processes to reduce audit risk and improve audit quality.
- Textual analysis for insights from language used to describe accounting and business outcomes and predictions.
- Cognitive technologies using artificial intelligence modeling to predict and classify phenomena.

These tools improve the efficiency and productivity of accountants. Regained worktime can be repurposed for more activities that add value, such as finding service opportunities and interpreting and communicating analysis results to stakeholders.

Data Mining and Smart Contracts

Some may know from personal experience that food sensitivity symptoms are easily diagnosed. However, the process of identifying which food(s) are associated with which symptoms can require disciplined testing over time. Health professionals often advise eliminating the most likely food sensitivity culprits, and systematically, over weeks and months, adding each back into the diet to identify specific food allergies.

Similar to the trial and error method of food allergy identification, data mining helps accountants systematically understand and test hypotheses while reducing some of the labor involved with data analysis. Additionally, smart contract software solutions allow business contracts to be securely negotiated, settled, and signed online in a fraction of the traditional time.

Data Mining

Increasingly sophisticated data mining tools are better able to handle the recent increases in data volume and data variety. These tools quickly diagnose errors and issues in the accounting data during the data preparation and exploration steps in the analysis stage. Data mining is also used to improve and defend management's estimations, assumptions, decisions, and assertions.

Data mining is the systematic practice of looking for issues or patterns occurring within or between data fields. Data mining lets us better understand data behavior and test expectations about data values, patterns, or relationships. The intelligence gained by data mining can lead to valuable operational changes:

- AIS accountants, IT auditors, and application programmers typically mine their data to find system delays or errors to identify where new policies, controls, or technologies can improve operations.
- Managerial accountants can use the information provided by data mining to increase sales revenues, save costs, or improve operations, such as improving credit by reducing how often accounts payable invoices are paid late.
- Tax accountants can use data mining to find tax deductions and tax credit qualifications for their clients.

Some of the leading data mining vendors include RapidMiner, Alteryx Analytics, SAS Enterprise Miner, Sisense, and MySQL. These tools range from those with easy-to-use, friendly interfaces to more complex tools that may require using programming languages, such as Python or R, to write the desired data mining and tests.

APPLYING CRITICAL THINKING 10.6: Reflect on Data Mining Results

You can improve data mining skills in a couple of ways **(Self-Reflection)**:

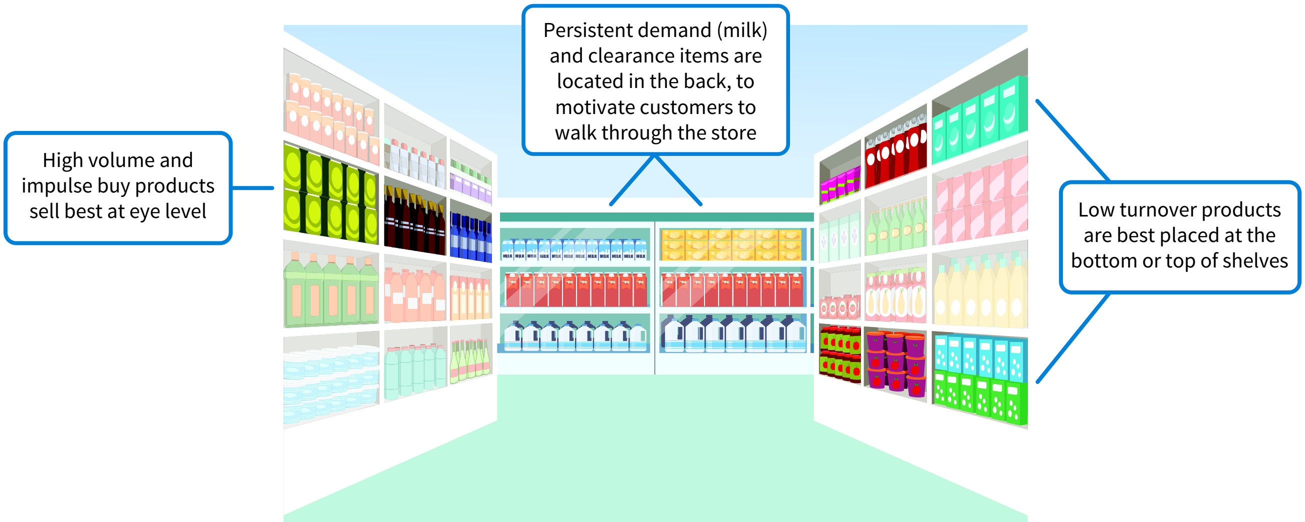
- Think about how the data value and veracity contribute to creating reliable intelligence and insights for the objective of the analysis. This awareness includes watching for results that might be random correlations with little or no business value.
- Estimate the potential data value that could be created by mining the data, and compare that value to the costs and risks associated with continuing current strategies. For example, looking for sales revenues to customers from states far from each specific store's location may help to focus efforts on fraudulent invoices.

Retail grocery stores, such as Wegmans, Publix, and Kroger, are credited as the first industry to create value from sales data mining. Empowered by point-of-sale cash register machines that integrated sales and AIS inventory data, supermarkets performed data mining to understand what store, aisle, and shelf locations were associated with increases in product sales:

- The first step was to create a categorical data field in their inventory tables which indicated store location (aisle number, front cap, back cap, store front table, register aisle), and shelf position (shelf number, and left or right side).
- The sales data with this location field was mined to learn which product locations were associated with high sales volume and which locations with low sales volume.
- Systematically and over time, each product's position in the store was moved to determine which positions resulted in higher sales for which products, as well as which products' sales were more resistant to location changes.

The next time you go to a grocery store you might notice some intentional product placement that is the result of mining store location data along with sales data. For example, temporary product promotions (e.g., strawberries during the spring) are best placed on the aisle end caps and store front display tables to capture customer attention (**Illustration 10.10**).

ILLUSTRATION 10.10 Product Placement Findings from Supermarket Data Mining



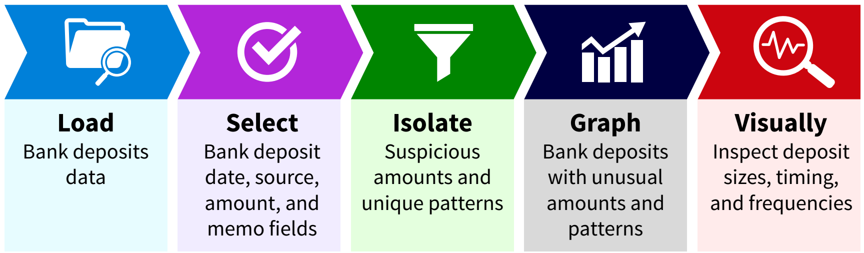
Let's consider an accounting data mining example. Assume you are hired as a consultant to help a law firm identify how to best use their limited professional staff resources. Your objective questions may include:

- Which legal service is the most profitable?
- Which service is performed most often?
- Which lawyer and paralegal team is currently earning the most client fee revenues for the law firm?
- Are any legal services being performed without being billed to clients?

Data mining tools can help gain insights to answer these questions. For example, you could use data mining to verify whether all services performed have corresponding invoices.

Data mining is perhaps the most common form of data analytics in accounting practice. It is useful for data preparation and data exploration. We can easily check for empty, incorrect, invalid, and duplicate data, and perform pattern identification tests, such as learning whether unusual logins indicate suspicious or unauthorized activity. Most data mining software displays the mining steps that have been performed on the data from left to right, which provides documentation for data verification procedures. **Illustration 10.11** illustrates a simple data mining process.

ILLUSTRATION 10.11 A Simple Data Mining Process for Identifying Suspicious Bank Deposits

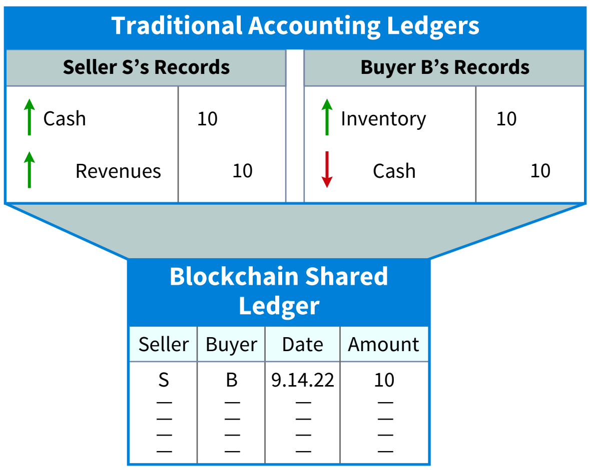


Smart Contracts

A business contract is created whenever two parties, such as buyers and sellers, agree on the consideration (usually money) that one party will provide to the other party in exchange for goods or services. Business contracts can range from simple exchanges to complex agreements with many stipulations. Contract review is important for proper GAAP classifications, such as revenue recognition or lease treatment. Historically, contract content was not standardized, which meant that both parties had to conduct a detailed review of the contract to ensure proper accounting in their separate journals and ledgers.

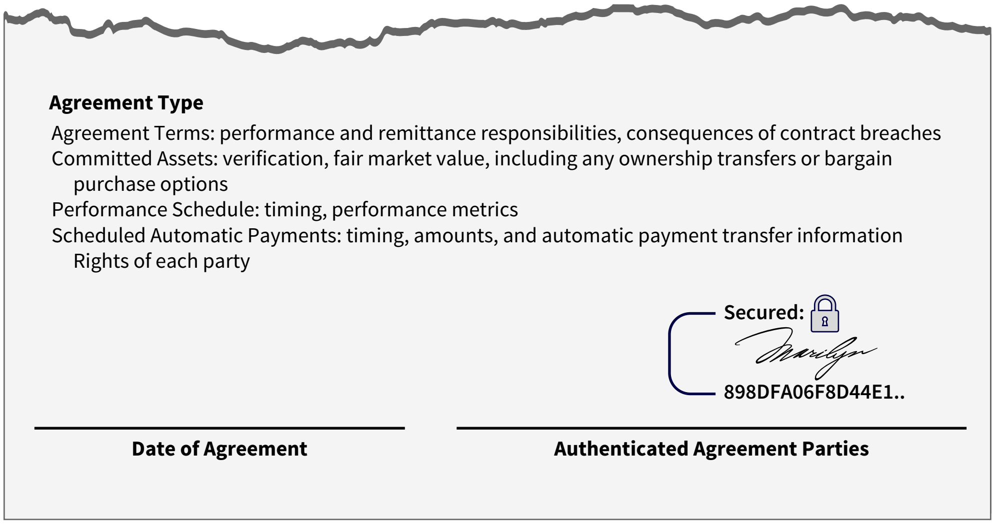
Blockchain technologies are a new online technology that creates one shared record of transactions between parties. These distributed ledgers are extremely secure and cannot be altered after an agreement has been reached. Blockchain technologies can be used for transactions with both digital and traditional currencies (**Illustration 10.12**).

ILLUSTRATION 10.12 Traditional Accounting Ledgers vs. Blockchain Ledgers



A specific and popular application of blockchain technology are **smart contracts**, which are secure online shared digital contracts. Smart contracts are now commonly used in healthcare, trading, lending, real estate, insurance, legal services, warranties, employment, and gaming contexts. **Illustration 10.13** shows some typical smart lease contract data fields.

ILLUSTRATION 10.13 Typical Smart Lease Contract Data Fields



Smart contracts have important advantages over traditional contracts:

- Contract templates provide organized, approved, and clearly labeled data field options, many of which are required by contract law or by GAAP rules for revenue recognition or lease categorization.
- Contract parties collaboratively develop contract specifics using secured online shared documents, reducing the time needed for contract negotiation, verification, and signing, while providing stronger internal controls. Contract options can be activated or deactivated as needed.
- All signed contracts cannot easily be changed by either party and all changes made are documented with red flags to all parties, which can eliminate many misunderstandings and disagreements.
- Smart contracts can automatically identify contract performance milestones, determine performance outcomes, and automatically trigger agreed payments between parties.

The standardized data fields make it easy for financial accountants to review and evaluate contracts and determine which GAAP rules are applicable. Auditors can easily verify the contract accounting. In both cases, time traditionally spent data mining contracts can be spent on other services that create value.

Process Automation and Process Mining

From doorbell cameras to robotic vacuums, smart sensors and programmable controllers are making some homes operate more efficiently and effectively. In the same way, business process automation tools are now performing routine tasks, freeing accountants to create value with critical thinking, problem-solving, and their resulting professional judgments.

While automated processes within software programs and equipment controllers have been available for decades, the recent development of across-program and across-equipment integrative tools have transformed many accounting tasks. In addition to greater performance accuracy and consistency, these new capabilities are producing activity logs that can be used to analyze and improve business processes.

Robotic Process Automation

Robotic process automation (RPA) is the set of software technologies that record a specific order of human keystrokes and mouse activity steps within or across digital applications. Once saved or recorded, these routines can be repeated by simply re-running the tool. RPA tools are used for stable processes where the steps of the process do not change over time, such as when financial analysts download the closing stock prices of the same set of companies every day.

Interactive, real-time performance dashboards are a great example of the value creation possible from RPA technologies. For example, the objective of a cost accountant for a manufacturing plant that produces breakfast cereals is to control product costs. Without RPA, controlling product costs might involve two hours daily of performing data analyses on the previous day's activity.

RPA technologies could recoup those hours with an automated routine which mines the production data every night and displays the results on the next morning's dashboard. The results could be interpreted each morning, which would make factory cost control efforts more effective and timely. **Illustration 10.14** illustrates a sample production activity dashboard that can result from implementing RPA.

ILLUSTRATION 10.14 Robotic Process Automation (RPA) Example Dashboard

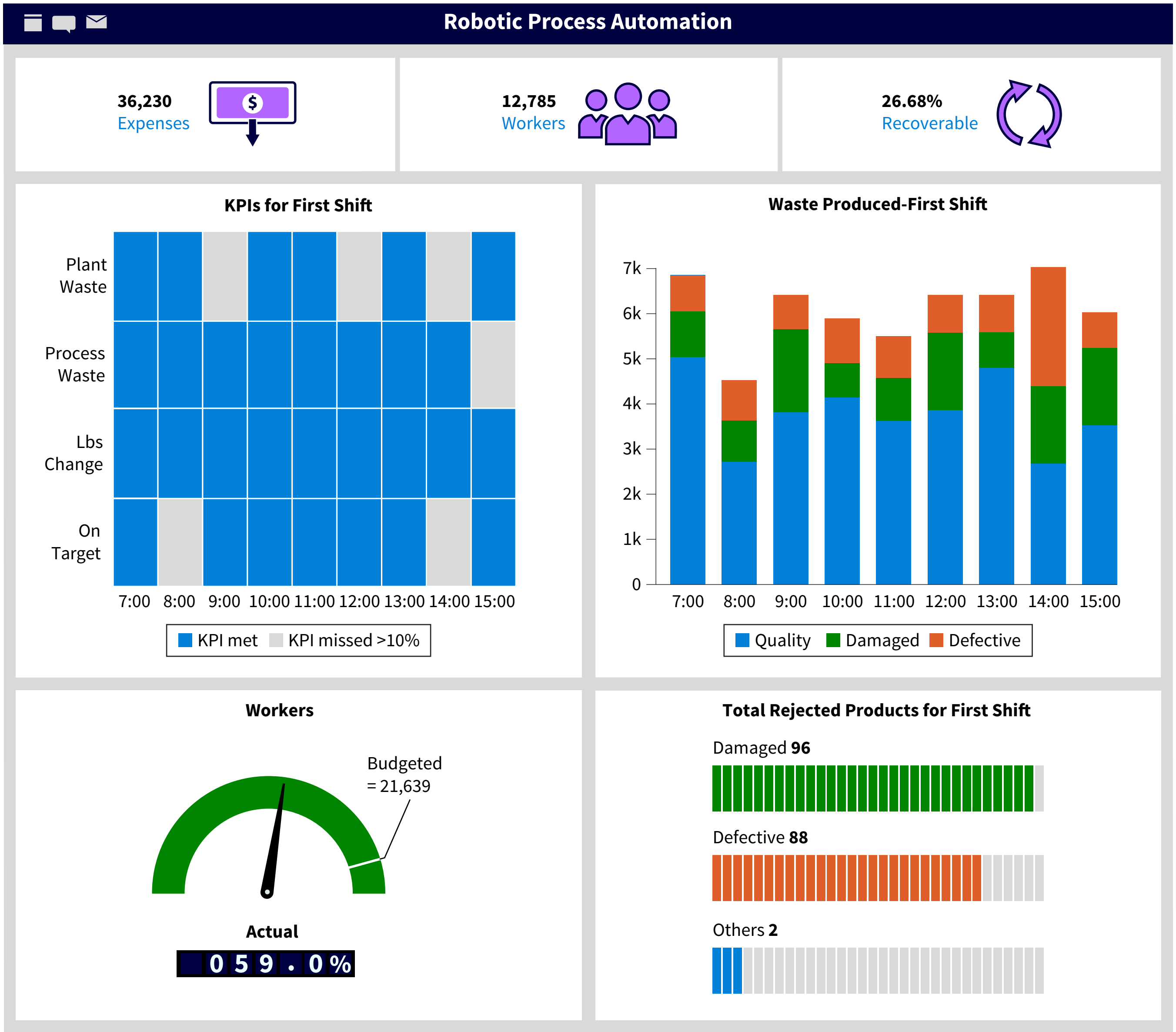


Illustration 10.15 summarizes more of the common accounting uses for RPA.

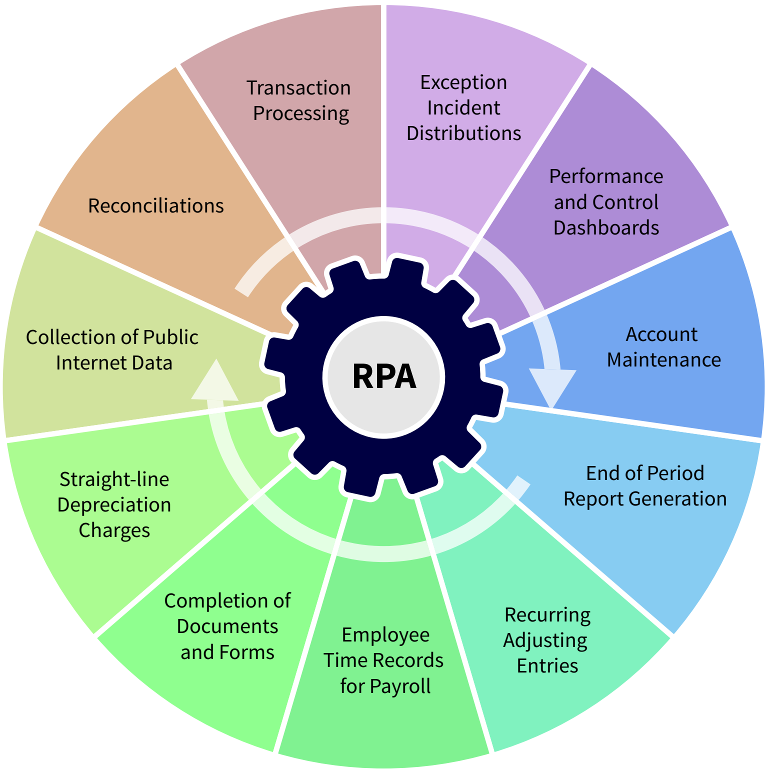


ILLUSTRATION 10.15
Accounting Tasks that Benefit from RPA

Professional Insight 10.1: How Can RPA Be Used in Financial Accounting?

The summer before the final year of her accounting program, Brigitte completed an internship where she gained valuable experience working for a financial analyst.

> **The first half of the summer I was gathering financial statement balances and totals for over 400 companies each day so I could calculate and compare 15 financial ratios.** Evaluating these ratio trends over time and comparing them to a competitor's ratios or to the industry average can provide insights about profitability, asset utilization, liquidity, and solvency.
>
> **About halfway through my internship, my boss suggested I learn how to use a routine process automation tool, UiPath, to write a routine to automatically gather the financial data.** My RPA routine opened the SEC data website and sequentially entered each company's code so the financial statement information would be displayed. My RPA script would then copy the data and paste it into an Excel worksheet, naming the sheet with that company's code.
>
> **I then learned how to write the RPA script for the calculation and comparison of all the financial ratios. What had previously taken me over six hours a day to perform was now running overnight so that I would have the information first thing the following morning.** Then, I could critically evaluate the ratios' meaning. I feel like my internship was much more valuable after automating the routine part of my day and spending my time learning the value of the information models I had created.

Process Mining

Process mining tools measure the timing and flow of data captured as business events occur. They help financial, managerial, and tax accountants better evaluate business process efficiency and effectiveness. Systems accountants also use process mining tools to evaluate AIS risks and control effectiveness. By describing where processes have more exceptions and delays, process mining also helps auditors adjust their audit plans to spend more time on the accounts and processes that have the greatest risk of financial statement errors and omissions.

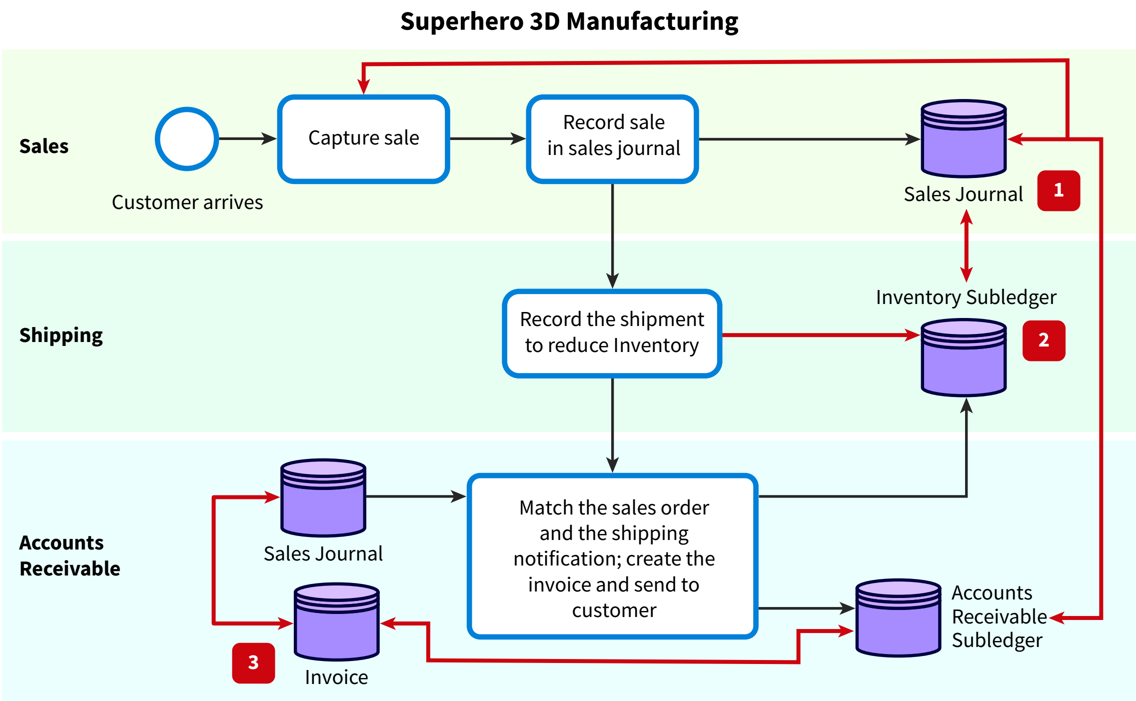
Process mining can improve, innovate, or control existing processes:

- Available data fields are typically extracted, joined, and transformed to create a process log.
- The process log records what happened where, who was involved, when it happened, and can capture which controls were implemented or skipped over.

Let's illustrate process mining with an example from a manufacturer of superhero figurines. **Illustration 10.16** depicts the three major events in their sales cycle:

- The sales department captures the sale and records it in the sales journal.
- The shipping department ships all sales notifications received from the sales department and reduces inventory.
- Accounts receivable matches sales orders and shipping notifications to create accurate invoices, which are sent to the customers.

ILLUSTRATION 10.16 Sales Cycle Example of Process Mining Tests



The red lines in Illustration 10.16 depict the specific tests to be conducted from the process mining, which can be expressed as these questions:

1. Are all sales captured reflected in the sales journal? What is the time delay between the capture and the recording of these sales?
2. Are all sales captured in the sales journal shipped? What is the time delay between the sales capture and the shipment to the customer? Are all shipments recorded as reductions in the inventory subledger? Are there any sales journal entries that are made and not shipped in the same accounting period? (GAAP rules state that revenue cannot be recognized until the sales transaction has been fulfilled.)
3. Do all invoices get posted to the accounts receivable subledger? How often and by how much are the sales journal entries different than the invoices? What is the time delay between the shipment and the invoice mailing to the customer?

The results may show that some captured sales are never shipped or do not ship in a timely fashion. Alternatively, there may be shipments that never turn into invoices, resulting in lost inventory. Finally, there may be differences between what is recorded in the sales journal and what invoice data is posted to the accounts receivable subsidiary ledger. To quickly show where and when the data variance happens for each of these questions, we could present process mining results as dashboard dials, as data tables, or as data visualizations.

Process mining results may also reveal violation of authorization or segregation of duties controls, such as the sales department both authorizing the sales capture and recording the sales journal entries. This insight would lead to changing the process so that accounts receivable is recording all sales transactions and sales events would not be recorded until after shipping and invoicing.

Two other examples help illustrate the value provided by process mining. In the AIS practice area, analyzing system access logs with process mining can ensure that system access for terminated employees is revoked in a timely manner. In managerial accounting, process mining can identify bottlenecks in business processes and establish relevant activity bases for activity-based costing systems. These examples document how process mining helps improve the internal controls over the accounting systems, reducing audit risk for both internal and external auditors.

Continuous Auditing

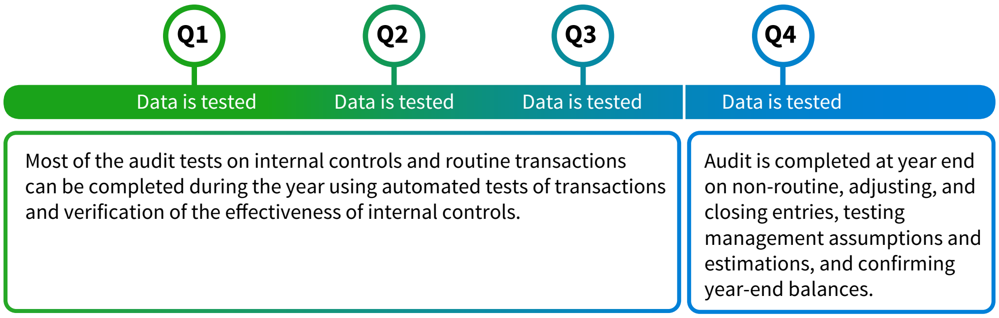
Have you ever wondered why auditors wait until after the year-end to perform their journal and ledger data tests under tremendous time pressure? Or why they use data sampling techniques to infer whether ledger balances are reasonable rather than evaluating the population of data?

Continuous auditing technologies are programmed modules embedded in an organization's AIS that evaluate transactions as they occur throughout the year:

- Routine transactions are typically captured and tested in automated routines for consistent processing.
- If transactions do not pass the tests to be categorized as a routine transaction, they are usually copied to an exception file to be evaluated by the external auditors at year end. Examples of these exceptions are invoice totals that are too high or too low for what has been sold or payroll cash disbursements which do not match up to an authorized employee.

This testing of the entire population of transactions helps auditors develop a more accurate assessment of the risks inherent to amounts reported on the financial statements. Continuous auditing tools reduce the workload at year end and provide faster audits with higher quality. **Illustration 10.17** illustrates continuous auditing modules employed at end of quarter throughout the year.

ILLUSTRATION 10.17 Quarterly Continuous Auditing Throughout the Year



The first users of continuous auditing were large banks and credit card companies. Both industries have millions of transactions occurring every day, which create a high data volume that is difficult to confidently sample by year end. More recently, companies in extractive and manufacturing industries have joined auditors in embracing continuous auditing technologies to support their internal applications.

Continuous auditing technologies also make audits more proactive rather than reactive. Accounting record issues and errors can be evaluated and corrected when they occur, increasing the integrity and reliability of internal data gathered after those corrections are made:

- Evaluating emerging data patterns earlier in the year can identify new tax strategies or areas where more internal controls are needed.
- Logging unusual internet use data points can identify cybersecurity risks earlier, which means that controls can be implemented.

- Automated evaluation of routine transaction data frees auditors to focus time and energy on audit process steps that contain more financial statement risk.

Tests of revenue recognition, inventory valuation, and cash flows are a few examples of areas where both internal and external auditors are using continuous auditing.

Textual Analysis

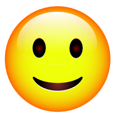
Textual analysis tools analyze word choices in footnotes, investor communications, blogs, social media postings, and other documentation, such as contract language for revenue recognition treatment. **Textual mining** tools evaluate word choice and usage to reveal insights about honesty, transparency, intent, and sentiments in communications by management, elected officials, and customers. Text counts and text patterns can be coded into categories of meaning, which are then analyzed along with numerical data for insights.

These tools make harvesting data efficient from sources such as financial footnotes and management emails to online customer feedback. Textual analysis can generate business intelligence, audit intelligence, and tax planning value. There are a few examples of text mining sources in the accounting profession:

- Lease contract language
- Annual reports, financial statement footnotes, and 10K or 10Q filings
- xBRL text data fields
- Auditor letters and reports
- Investor relation press releases
- Correspondence from regulators
- Tax planning and response letters

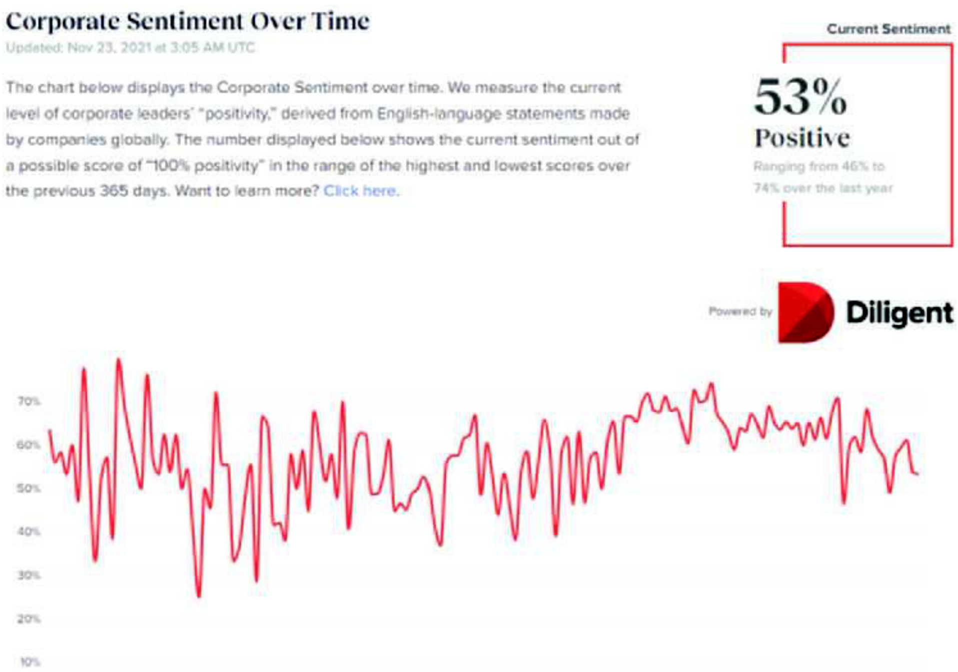
We can also use textual analysis on formal and informal social media communications by analysts, auditors, and regulators to categorize the favorability or pessimism in those communications (**Illustration 10.18**).

ILLUSTRATION 10.18 Optimistic and Pessimistic Word Categories: Annual Report Examples

Optimistic Terms	Pessimistic Terms
	
Applaud	Bleak
Favorable	Costly
Successful	Troubled
Positive	Unstable

Diligent Institute is the leading technology solution designed to improve board of directors' fiduciary responsibilities to oversee management decisions. They create financial and textual analysis dashboard visualizations to focus board members' attention on the areas of the organization with the greatest risks. For example, they have used textual analysis to show corporate board members' sentiment rating over time. These results show how optimistic corporate board members are about the upcoming opportunities and performance possibilities in their organizations (**Illustration 10.19**).

ILLUSTRATION 10.19 Diligent Institute's Board Member Sentiment Tracker



Source: www.diligentinstitute.com/sentiment-tracker/

Textual analysis that evaluates the use of passive versus active verbs and the ratio of positive words to negative words, often called the polarity of the communication, helps to categorize communications. These categories can then be analyzed in combination with financial statement totals or ratios to better predict future earnings, bankruptcy risk, or stock prices.

Cognitive Technologies

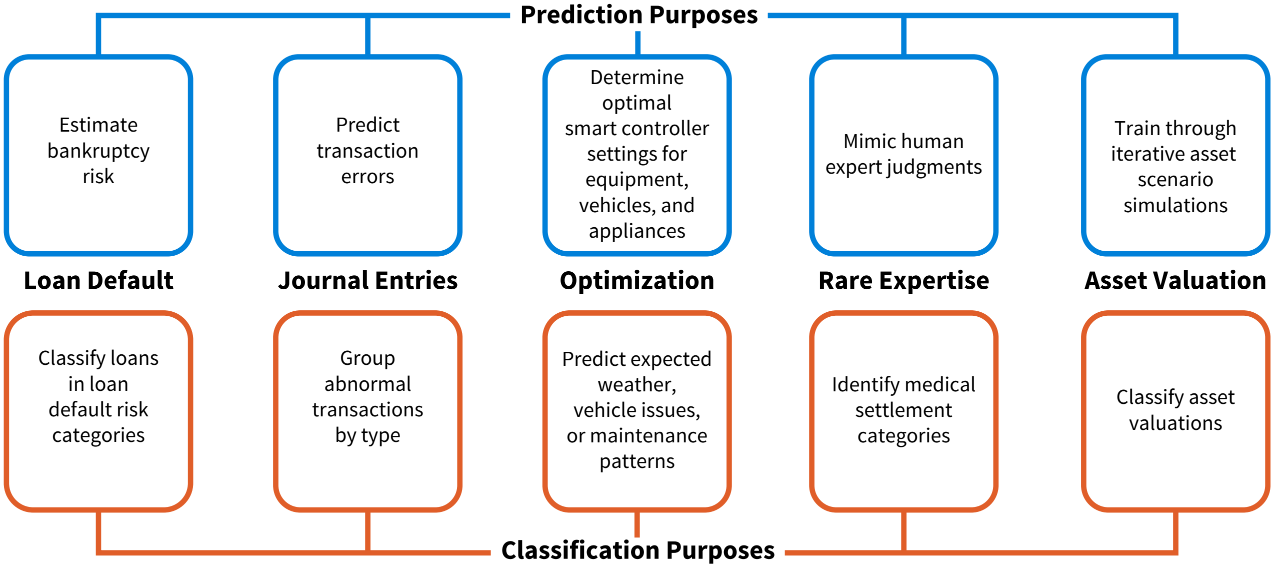
If you have interacted with Apple devices, then you have witnessed the power of Siri, the virtual assistant feature of Apple's operating system. Siri is a user-friendly artificial intelligence tool programmed to answer most questions and research requests. **Cognitive technologies** are artificial intelligence technologies that use a human expert judgment mimicking internal algorithm. Accountants typically use AIS data to train their models for accuracy for diagnosing and predicting objectives, and then adjust their model as new data is added. Cognitive technologies are used in many fields, including training, production, farming, banking, retail, healthcare, tourism, and utilities, and in a variety of business functions within organizations. They can range from simplistic decision models to dynamic, complex models.

Complex cognitive tools are also called artificial neural networks, machine learning, and expert systems. These models are programmed to accurately perform data classification tasks for diagnostic or prediction objectives. For the model to be accurate, the underlying phenomena variables should behave in stable or consistent patterns.

Cognitive technology tools perform best in stable environments because they use old data to define a complex mathematical algorithm which will, over time, produce an expert decision. The most advanced models, called adaptive cognitive technologies, can adjust their underlying models with every new data point, but these changes are typically very small, as the models were trained on very large data sets.

Illustration 10.20 lists a few common uses of cognitive technologies in accounting.

ILLUSTRATION 10.20 Cognitive Technology in Accounting



The accounting profession is embracing many valuable applications of cognitive technologies across all professional practice areas. The Committee of Sponsoring Organizations (COSO), which provides accounting internal control and risk management models, expects these tools to provide value through:

- Increased security and internal control.
- Improved cost reductions, operational productivity, and process quality.
- Predictive analytics.
- Innovation through iterative testing.

APPLYING CRITICAL THINKING 10.7: Recognize Risks with Cognitive Technologies

Using cognitive technologies is not risk-free. For example, if a classification model incorrectly rejects a loan application due to a data error, then the lending institution could lose business. COSO suggests considering the following **(Risks)**:

- Potential biases by the expert who developed the cognitive technology.
- High potential for errors from data and interpretation.
- Changing laws and regulations that may have been overlooked in the model.
- Changing underlying conditions or ranges of values that may make your model less predictive.
- Vulnerabilities to security attacks.
- Negative reactions from employees, customers, and other impacted stakeholders.

COSO also recommends cognitive technology policies to increase internal controls:

- Establish governance structure for all cognitive technology tool applications.
- Develop a risk management strategy.

- Regularly assess the risks of the organization's cognitive technology applications.
- Evaluate the cognitive technology applications as a portfolio rather than individual assets.
- Document and internally share the risk appetite and risk response commitments associated with cognitive technology use to increase transparency.

Apply It 10.2
Use Process Automation to Analyze Financial Statements

Financial Accounting **Data** The macro record option in Excel illustrates the power of RPA for financial statement analysis. Try using it to analyze accounts receivable liquidity for a selection of leading outdoor gear companies.

You have developed an objective question: Which of the leading outdoor gear industry companies had the best liquidity with regards to accounts receivable? The following data is provided for each company for the years 2014-2019:

- Income statement: Net sales revenues, cost of goods sold, gross profit, and net income.
- Balance sheet: Year-end accounts receivable and inventory balances.

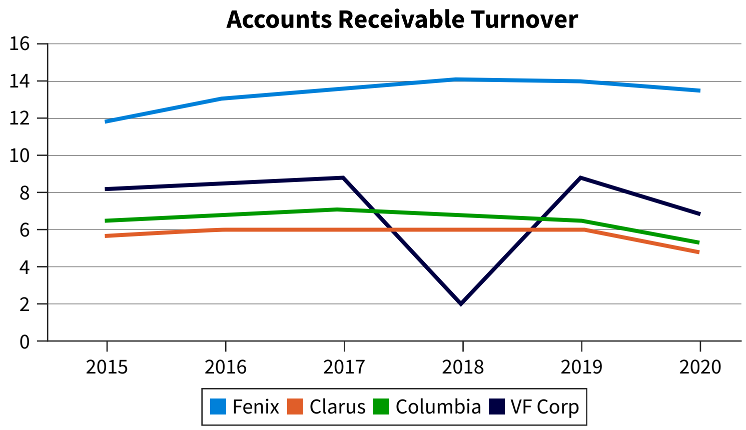
Record one macro routine on the first worksheet to calculate accounts receivable turnover, and then repeat the macro on the subsequent workbook sheets that contain the financial statement data from other companies.

SOLUTION

Ratio Values and Graph from the Comparison worksheet.

Comparison of Accounts Receivable Turnovers

	2015	2016	2017	2018	2019	2020
Fenix	11.9	13.1	13.6	14.1	14.0	13.5
Clarus	5.7	5.9	5.8	5.9	5.9	4.9
Columbia	6.5	6.7	7.1	6.9	6.5	5.3
VF Corp	8.3	8.6	8.8	2.1	8.7	6.9



Fenix Outdoor has the best accounts receivable liquidity throughout this six-year period.

Building and Using the Excel Macro

1. Open the Apply It 10.2 data file. Your platform may ask you to enable macros and to make this file a trusted file. Select **Yes** for both.
2. Go to the first sheet for Fenix Outdoor and select the **View** tab. Select **Macro** (last option on right). Select **Record** and name your new macro "AccRecTurnover."
3. Place the cursor on cell C18. Type the formula "=C7/((sum(B13:C13))/2)" and click **Enter**.
4. Copy this formula to D18-H18. Remove the adjacent cell error by right clicking on the error icon to the left of this formula and select **Ignore error**.
5. Go back to **View Macro**, and stop recording.

6. Next, go into each remaining company data sheets and click Cell C18. Select **View** > **View Macros** > **AccRecTurnover** > **Run**. Watch as the macro performs the ratio calculations for each company, saving you from having to re-create the steps for each company. Remove the adjacent cell error on each sheet by right clicking on the error icon to the left of this formula and selecting **Ignore error**.
7. Now, open the Comparison worksheet. All the accounts receivable turnover results linked to this comparison sheet should be visible. Highlight all the data cells, B3:H7.
8. Under top menu, select **Insert**, and add a line chart.
9. Double click on the chart title and change it to "Accounts Receivable Turnover." Compare results.

10.3 How are Data and Analyses Developments Impacting the Professional Practice Areas?

LEARNING OBJECTIVE 3

Demonstrate how data and technology developments are adding value to professional practice.

Emerging data, analyses, and technologies are significantly impacting each area of professional practice. Accounting tasks are more accurate and completed in less time, freeing accountants to focus on using critical thinking and creative ways to add value to their organizations.

Accounting Information Systems

Accounting information systems professionals manage the design, implementation, and evaluation of the accounting system transaction processing and business intelligence information. Many new data options are improving data volume, variety, veracity, and value for AIS users. New analysis tools discussed here help systems accountants improve their AIS data analyses:

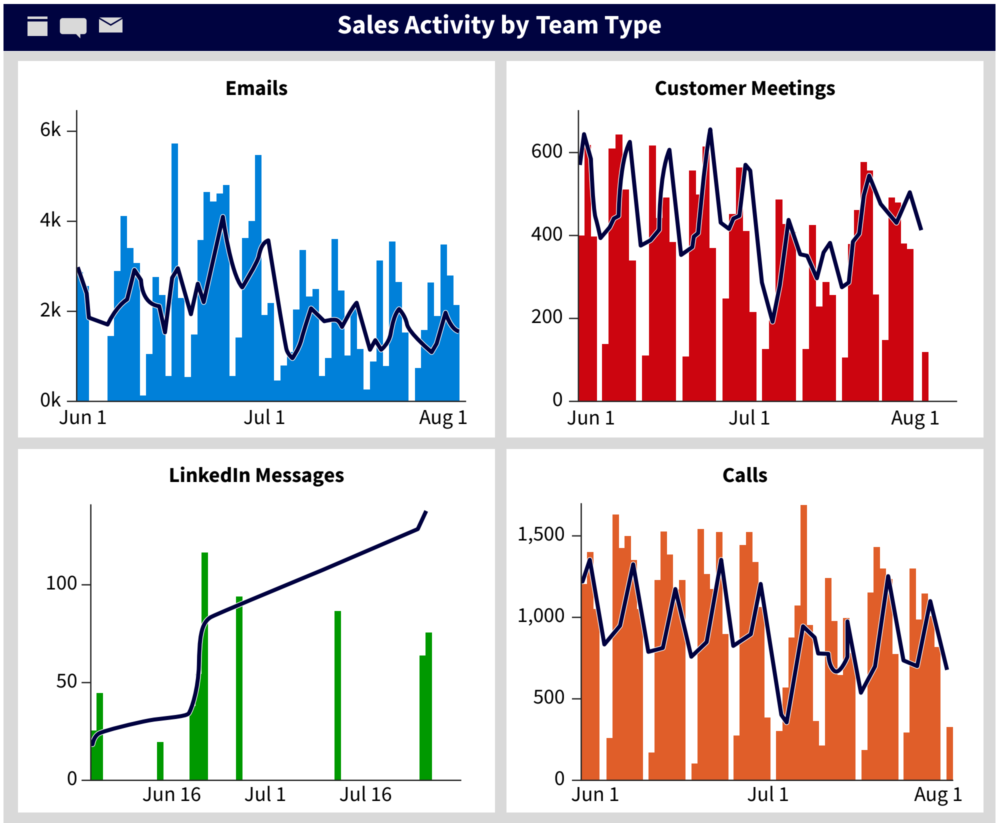
- Detecting unauthorized intrusion (data mining, process mining).
- Transaction cycle throughput, internal controls effectiveness, delays, and errors (data mining, process mining, continuous auditing).
- Artificial intelligence supports and supplements scarce IT experts (cognitive technologies).
- Routine data and system maintenance (RPA).
- Automating system updates (RPA).
- Automatic performance reporting for department managers (RPA).

With these new data and technology options, systems accountants can consider alternative data capture, storage, processing, controls, and performance evaluation methods:

- One possibility is adopting the capacity and value offered by cloud data services.
- Another is using internal continuous auditing and process mining to evaluate system performance and security issues.
- They can also identify processes where cognitive technologies and routine process automation can allow employees to spend more time interpreting information and adding value to their organizations.

For example, systems accountants can help marketing and sales departments by creating visualizations of the technologies being used by their key managers (**Illustration 10.21**).

ILLUSTRATION 10.21 AIS Analytics Value for Marketing and Sales Functions



Auditing

Both internal and external auditors also using many of the emerging data and technology options to increase the quality of audits while reducing their costs. RPA, cognitive technologies, process mining, and continuous auditing are improving analytical testing of controls and substantive testing since an entire population of transactions can be analyzed.

New combinations of data, information models, and graphics make it easier to understand a client's business and data (e.g., dashboard visualizations such as Illustration 10.15), and can be used to communicate and audit's "story" to management and boards of directors.

AIS audit trail log data (user ID, date, and time stamp) is helpful for verifying segregation of duties. Continuous auditing also has advantages:

- Reduced audit sample risk, as inferences from limited samples are reduced or eliminated from the process.
- Auditors can focus on prioritized risks in their clients' data, such as management's assertions and estimates, and increase their audit quality consistency across clients.
- Audits can be finished closer to the fiscal year end date, which will increase the value of the client's financial statement information in the capital markets.

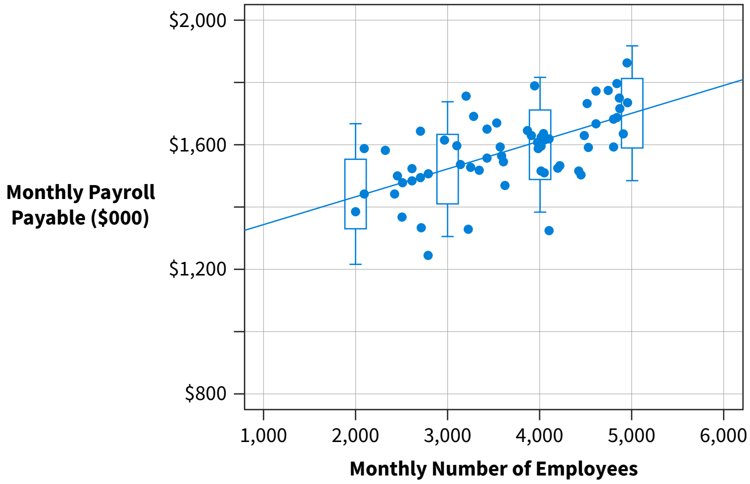
Audit firms can also develop their own intellectual property intelligent assets as they create confidential tools for their firms. However, there may be audit risks associated with these emerging data analytics options, such as client AIS and auditor tool compatibility, and issues with data security and privacy. For example, audit risk is impacted if clients' systems are not compatible with new audit tools, if the client uses an integration of different systems patched together, or if the data integrity from the clients' system is low. In these cases, the tools may not retrieve enough data or the appropriate data for the auditors' evaluations.

Auditors are performing a variety of their tasks using the tools discussed in this chapter:

- Classifying unusual transactions (cognitive technologies).
- Performing risk level assessments (continuous auditing and cognitive technologies).
- Selecting better samples for substantive testing (blockchain technologies and data mining).
- Performing analytics on selected data for substantive testing (process mining).
- Testing transactions for routine qualities (continuous auditing modules, cognitive technologies, and data mining).
- Monitoring and evaluating internal controls (process mining).
- Evaluating ongoing concerns (cognitive technologies).

Auditors often use analytics to improve judgments about whether recorded account balances are reasonable, which helps them improve their substantive test sample selections (**Illustration 10.22**).

ILLUSTRATION 10.22
Visualization used by Auditors for Reasonableness of Payroll Expenses.



Financial Accounting

Financial accountants are also using these new data, analysis, and technology developments for both their GAAP compliance responsibilities and their assessments and predictions of business value, risk, and future capital needs. Some of the more common application areas of these technologies include:

- Reconciliations (RPA).
- Temporary account closures (RPA).
- Routine consolidations (RPA and process mining).
- Accounting period end closing entries (RPA).
- Footnote preparation (textual analysis).
- Statutory and regulatory report content generation (cognitive technologies).
- Proforma financial statement generation (data mining and pattern detection).
- Scanning of changes in external environment (textual analysis).
- Data patterns and relationships to explain stock market valuation changes (data mining, blockchain, and continuous auditing).

Routine process automations are a natural fit for financial accounting. Stable, iterative revenue, expenditure, payroll, production, and financial accounting processes can easily be captured and then reused with recorded macros. Additionally, process mining helps evaluate the efficiencies, effectiveness, and controls in each accounting step (see Illustration 10.16).

Cognitive technologies can be used to both predict financial outcomes and classify business option risks. Textual content is analyzed for regulatory implications and earnings expectations communication. Finally, many CFO's can now monitor their financial ratio performance real-time using live dashboards tied to their accounting systems (**Illustration 10.23**)

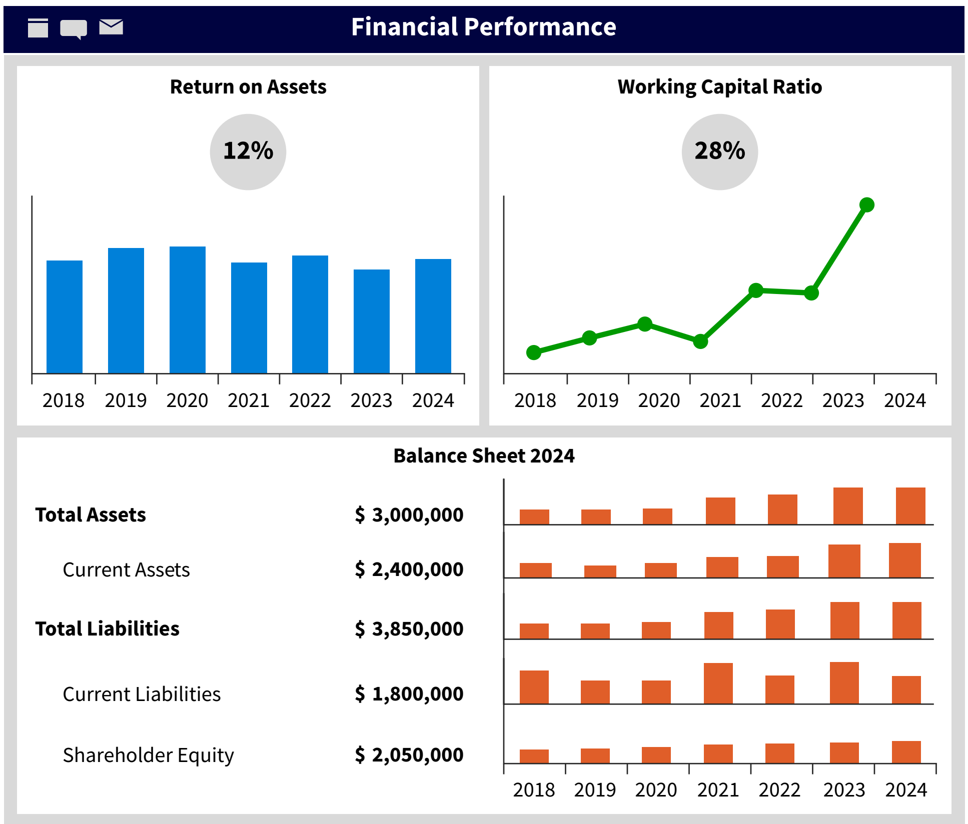


ILLUSTRATION 10.23
Sample Dashboard for Financial Accountants

Managerial Accounting

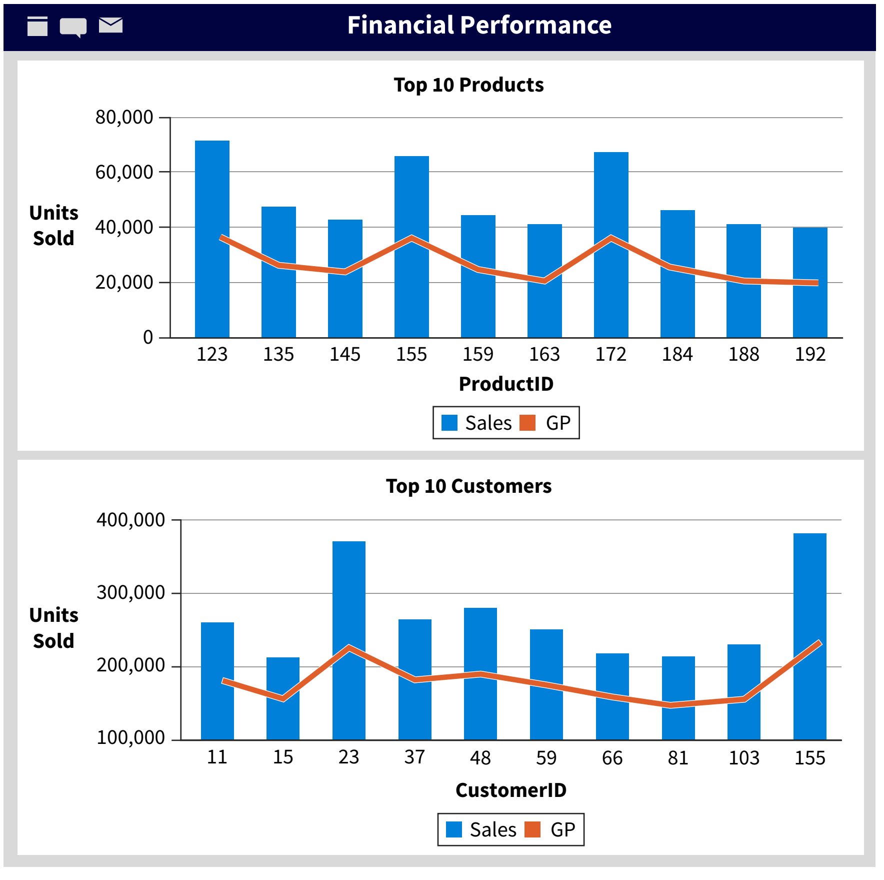
Managerial accountants are not tied to the GAAP data restrictions or the IRS data rules, allowing them more freedom and creativity to solve problems and identify opportunities than the more regulated practice areas:

- Managerial accountants have traditionally used broad sets of operational and market data to find the most effective least expensive solutions for their organizations.
- As confidential advisors to the management team, they tend to quickly embrace new data analyses opportunities, for descriptive, diagnostic, predictive, and prescriptive objectives in budgeting, costing, pricing, investment, or controlling tasks.
- These new analysis tools are helping to improve the performance of functional departments, such as human resources, marketing, sales, and inventory management.

The new data options let managerial accountants add value by helping them ask better questions, accurately identify priority issues, and develop intelligence and solutions to support their management team's decision-making. Because smart devices and social media are increasingly used by organizations, managerial accountants are managing data with greater volume, variety, and velocity than before, and they must control data veracity by properly capturing, processing, and storing data.

Several new technologies are helping managerial accountants improve processes and add value to their organizations. RPAs can quickly create daily performance dashboards across organizational functions, cognitive technologies forecast sales and production levels, data and process mining help optimize operational control, and blockchain technologies are capturing resource transfers within the organization, such as inventory transfers across warehouses. Managerial accountants can keep track of their areas of responsibility using the tools discussed in this chapter to create visualizations for dashboards (**Illustration 10.24**).

ILLUSTRATION 10.24
Sample Dashboard for Managerial Accountants



Process mining is helping managerial accountants identify business process areas where costs can be reduced:

- Mining transactions for increased travel expense, miscellaneous expense, or overhead cost account to control costs.
- Cost savings in general, including reducing headcounts, are easier to identify with process mining.

Other technologies have applications, as well:

- Improving decision-making by making more types of data accessible for analysis.
- Discovering new business insights in revenue, expenditure, and operational processes (data mining, process mining, and cognitive technology).
- Increasing employee productivity with useful dashboards and automations of routine task completion (RPA).
- Enhancing product offerings through innovation desired by customers (textual analysis of social media postings).

Managerial accountants are also faced with new challenges, many of which include ethical considerations about digital data privacy and security (see Illustration 10.7 and Applying Critical Thinking 10.5). Another challenge is data veracity, since dirty data can produce incorrect analysis results which can negatively impact decision-making. Properly preparing data catch and repair corrupt, incomplete, inaccurate, or inappropriate data formats that improve data analyse results' reliability.

Finally, managerial accountants might miss opportunities in data that they do not use in their analyses. Dark data refers to the available operational data that are not used by an organization. The related costs of data capture, management, and storage can be significant, and if the data are not being used, then the possible business intelligence value may be reduced.

Tax Accounting

While initially the slowest area of accounting practice to use data analytics, tax accountants are now moving quickly toward using many new data and analysis technologies:

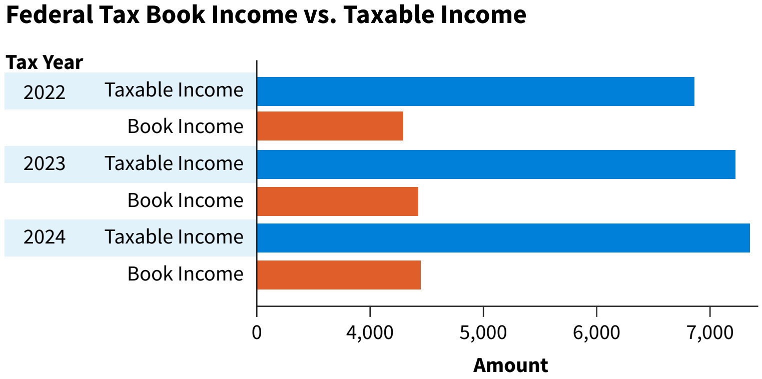
- Cognitive technologies are heavily utilized for compliance testing and for the evaluation of multi-jurisdictional sales and use taxes.
- Textual analysis is increasingly used to better understand and interpret new tax law in the unstructured data posted on the internet.
- Process tools help tax accountants better understand which processes in an organization or client's portfolio are driving their tax liabilities, improving the value of tax planning significantly.

Data pattern documentation has been a long-held tradition for tax treatment choices. For example, consider when a tax accountant must explain the "tax shield" power of LIFO cost of goods sold valuation in inflationary periods to board members:

- If the tax accountant simply states that the LIFO tax shield defers income tax liabilities to a later taxable period without explaining why, the meaning of why this is the case may be lost.
- But, if that accountant additionally illustrates and explains how a LIFO adoption and a LIFO adoption for inventory valuation on the balance sheet creates a stronger current cash flows relationship with cost of goods sold on the income statement, the board members can more easily follow how current tax liabilities and cash outflows would be reduced by the company's "tax shield."

Both tax planning and compliance practice areas are demanding more efficient processes, effective planning, accurate compliance, and increased tax intelligence. These new technologies are delivering this by analyzing greater data volume and data variety. RPA is providing efficiencies and decreasing tax function costs. Data mining visualizations can identify tax liability drivers. Cognitive technologies provide accurate tax liability predictions, as well as proper transfer pricing and tax treatment classifications, especially in assessing complex multi-entity and multi-jurisdictional tax liability contexts (**Illustration 10.25**).

ILLUSTRATION 10.25 Tax Accountants Use Visualizations to Guide Their Judgments



Professional Insight 10.2: How are New Technology Tools Used in Tax Accounting?

Brandi recently graduated with her Master of Accountancy and is working for a regional tax practice. She explained how she uses data analytics in her daily work.

> **Because different countries have different value-added, sales, and use tax policies and rates, companies have to be very strategic determining which parts of their operations are performed in which countries. Several new tools have helped me perform this work.**
>
> Transfer pricing refers to the prices at which we sell resources from our business units or subsidiaries in one tax jurisdiction to our companies in another tax jurisdiction for the purpose of minimizing tax liabilities. The different processing steps must be considered, from raw materials to finished goods, to determine where our company processes should be located.
>
> **I use a variety of advanced analysis tools in my daily tasks. I describe data with a data mining tool, and I gain insights about the risk in the data from process mining tools.** I really enjoyed the freedom to help my company figure out the best country to create, further process, or assemble these resources. The objective is to operate legally while minimizing operational costs and tax liabilities. Before these process mapping technologies, such as Alteryx and Celonis, were available, we had to build individual Excel files for each possible operational combination between countries. Creating these spreadsheets was time consuming, they poorly documented the original purpose, and it was difficult to integrate and evaluate the information across these spreadsheets.
>
> **I used robotic process automation to retrieve and enter the data for the spreadsheets, which has saved me more than a dozen hours every week. A particular advantage of moving our work to newer technologies such as PowerBI and Alteryx, is that automatic documentation is capturing each step of our workflows and our data manipulation processes alike.**

Analyses that integrate data variety and data mining visualizations are helping tax accountants to better understand when their organizations or clients are approaching the boundaries of acceptable tax categories. The technology advances reviewed in this chapter are helping tax accountants drill down to supply chain events that are driving tax liabilities, advancing real-time tax advice potentially saving organizations and clients significant cash resources.

Apply It 10.3
Match Professional Practice Areas to Analyses and Technologies

Match how the following data analyses and technologies are used to at least one professional practice area's acronym. Accounting information systems (AIS), auditing, financial accounting, managerial accounting, and tax accounting.

Accounting Area	Data Analyses/Technology Uses
	Forecasting classifications of new customers into post-sale interaction categories with machine learning.
	Evaluating cloud data usage patterns with process mining.
	Standardizing equipment lease contracts with blockchain smart contracts.
	Documenting segregation of duties in the expenditure cycle with process mining.
	Optimizing taxable transfer prices across global facilities with machine learning.
	Communicating operational strategic metric progress to management with storytelling.
	Closing revenue, expense, gain, loss, and dividend accounts at year end with RPA.
	Controlling unauthorized system access with process mining on log-in data.
	Verifying accounts receivable balances with fulfilled invoice data using process mining.

SOLUTION

Accounting Area	Data Analyses/Technology Uses
Managerial accounting	Forecasting classifications of new customers into post-sale interaction categories with machine learning.
AIS	Evaluating cloud data usage patterns with process mining.
Financial accounting	Standardizing equipment lease contracts with blockchain smart contracts.
Auditing	Documenting segregation of duties in the expenditure cycle with process mining.
Tax accounting	Optimizing taxable transfer prices across global facilities with machine learning.
Managerial accounting	Communicating operational strategic metric progress to management with storytelling.
Financial accounting	Closing revenue, expense, gain, loss, and dividend accounts at year end with RPA.
AIS	Controlling unauthorized system access with process mining on log-in data.
Auditing	Verifying accounts receivable balances with fulfilled invoice data using process mining.

Chapter Review and Practice

Learning Objectives Review

1 Describe data trends impacting the accounting profession.

Trends in data volume, variety, velocity, veracity, and value are creating both opportunities and challenges across accounting areas:

- Data volume refers to amount of data used in analyses. The increased volume of data generated and collected by organizations have led to new developments in data storage. Cloud data services offer organizations secure and easily expandable data storage, expert data management, hardware and software, and IT expertise. Private clouds serve a single organization or a set group of non-competing organizations. Public clouds serve many different businesses. Hybrid cloud storage have both private and public virtual server sections.
- Data variety refers to the diversity of data structures and measurement scales in the data we want to analyze. Accountants have traditionally selected and analyzed structured data, but are now combining structured data with unstructured data for analysis, using pictures and text and GPS data, for example.
- Data velocity is the speed at which new data points are generated. The higher the data velocity, the faster data volume grows. High velocity data can come from both internal or external sources.
- Data veracity is the reliability of data. Veracity depends on three things: if the data are expected to provide insights for the analysis, how accurate the data must be to generate useful insights, and how complete the data must be to generate useful insights.
- Data value refers to the additional benefits the data provides toward the objective of an analysis.
- There are also ethical considerations when considering the storage and use of data. Accountants must consider data privacy laws and regulations when working with, storing, and sharing data.

2 Explain how technology developments are impacting data analysis in accounting.

Accountants can efficiently analyze more data than ever before using data analysis tools:

- Data analyses can produce new intelligence for improving how resources and processes can increase operational performance, competitiveness, and achieve financial objectives.
- Data mining is the systematic process of looking for issues or patterns occurring within or between data fields. Data mining is a common form of data analytics in accounting practice. Most data mining software automatically documents the mining steps, which is useful for data and analyses verification.
- Blockchain technologies create one shared, secure, and difficult to change record of transactions between parties. This technology creates smart contracts, which are secure online shared digital contracts.
- Robotic process automation software automates routine, stable processes that can then be easily repeated.
- Process mining tools measure the timing and flow of data captured as business events happen. These tools help accountants better evaluate business process control, efficiency, and effectiveness.
- Continuous auditing technologies are programmed modules embedded in the AIS that evaluate transactions as they occur throughout the year, reducing the year-end work.
- Textual analysis tools help accountants analyze word choices in footnotes, investor and management communications, and other documents, such as legal contracts.
- Cognitive technologies are artificial intelligence technologies that mimic human expert judgments.

3 Demonstrate how data and technology developments are adding value to professional practice.

- Accounting information systems accountants can add value using AIS system log data and cognitive technologies, continuous auditing, data mining, process mining, and robotic process automation.
- New technologies can increase audit quality and reduce audit time and costs. RPA, cognitive technologies, process mining, and continuous auditing are improving analytical testing of controls and testing of entire populations of data, rather than just inferential samples, are creating new audit value.
- Financial accountants use new technologies in both their GAAP compliance responsibilities and for assessing and predicting business value, risk, and future capital needs. RPA, process mining, cognitive technologies, and textual analyses are all being used in the financial accounting function.
- New data options and technologies are helping managerial accountants better identify priority issues, ask better questions, use data with greater value, and develop intelligence solutions to support management decision-making.
- Tax accountants use cognitive technology for compliance testing, textual analysis to understand and interpret unstructured data, and process tools to better understand processes that affect tax liabilities.

Key Terms Review

Blockchain technologies 10-16
Cloud data services 10-3
Cognitive technologies 10-23
Continuous auditing technologies 10-21
Data mining 10-14
Data value 10-8
Data variety 10-5
Data velocity 10-6
Data veracity 10-7
Data volume 10-2
Hybrid clouds 10-4
Private clouds 10-3
Process mining tools 10-19
Public clouds 10-4
Robotic process automation (RPA) 10-17
Smart contracts 10-16
Textual analysis 10-22
Textual mining 10-22

How To Walk-Throughs

HOW TO 10.1 Create a Comparative Income Statement in Excel

How To

Illustration 10.6 demonstrated how considering the value of data can provide useful insights when deciding whether to drop a net loss-producing product line of computer keyboards (Product B). As a reminder, the segment contribution margin calculation was more valuable than the contribution margin calculation because it considered the traceable fixed costs which could be avoided if the product line were dropped.

What You Need: **Data** The How To 10.1 data file.

STEP 1: Open the Excel file and review the data field columns extracted from this company's AIS database. The retrieved data has summarized monthly totals for units sold, price, and cost information for the 2025 year for two product lines, A and B (**Illustration 10.26**).

ILLUSTRATION 10.26 Data Set Columns

1

	A	B	C	D	E	F	G	H	I	J	K	L
1	Year	Month	Product	SumUnits Sold	Price	VCunit	Traceable FCunit	Common FCunit	Total Revenues	Total DirectVC	TotalApplied TraceableFC	TotalApplied CommonFC
2	2025	January	A	600	$ 60.00	36.00	17.50	3.50	$ 36,000.00	$ 21,600.00	$ 10,500.00	$ 2,100.00
3	2025	January	B	1,710	$ 100.00	80.00	18.75	3.50	$ 171,000.00	$ 136,800.00	$ 32,062.50	$ 5,985.00
4	2025	February	A	700	$ 60.00	36.00	17.50	3.50	$ 42,000.00	$ 25,200.00	$ 12,250.00	$ 2,450.00
5	2025	February	B	1,640	$ 100.00	80.00	18.75	3.50	$ 164,000.00	$ 131,200.00	$ 30,750.00	$ 5,740.00
6	2025	March	A	760	$ 60.00	36.00	17.50	3.50	$ 45,600.00	$ 27,360.00	$ 13,300.00	$ 2,660.00
7	2025	March	B	1,360	$ 100.00	80.00	18.75	3.50	$ 136,000.00	$ 108,800.00	$ 25,500.00	$ 4,760.00
8	2025	April	A	740	$ 60.00	36.00	17.50	3.50	$ 44,400.00	$ 26,640.00	$ 12,950.00	$ 2,590.00
9	2025	April	B	1,270	$ 100.00	80.00	18.75	3.50	$ 127,000.00	$ 101,600.00	$ 23,812.50	$ 4,445.00
10	2025	May	A	820	$ 60.00	36.00	17.50	3.50	$ 49,200.00	$ 29,520.00	$ 14,350.00	$ 2,870.00
11	2025	May	B	1,200	$ 100.00	80.00	18.75	3.50	$ 120,000.00	$ 96,000.00	$ 22,500.00	$ 4,200.00
12	2025	June	A	900	$ 60.00	36.00	17.50	3.50	$ 54,000.00	$ 32,400.00	$ 15,750.00	$ 3,150.00
13	2025	June	B	940	$ 100.00	80.00	18.75	3.50	$ 94,000.00	$ 75,200.00	$ 17,625.00	$ 3,290.00
14	2025	July	A	980	$ 60.00	36.00	17.50	3.50	$ 58,800.00	$ 35,280.00	$ 17,150.00	$ 3,430.00
15	2025	July	B	860	$ 100.00	80.00	18.75	3.50	$ 86,000.00	$ 68,800.00	$ 16,125.00	$ 3,010.00
16	2025	August	A	1,040	$ 60.00	36.00	17.50	3.50	$ 62,400.00	$ 37,440.00	$ 18,200.00	$ 3,640.00
17	2025	August	B	720	$ 100.00	80.00	18.75	3.50	$ 72,000.00	$ 57,600.00	$ 13,500.00	$ 2,520.00
18	2025	September	A	1,220	$ 60.00	36.00	17.50	3.50	$ 73,200.00	$ 43,920.00	$ 21,350.00	$ 4,270.00
19	2025	September	B	650	$ 100.00	80.00	18.75	3.50	$ 65,000.00	$ 52,000.00	$ 12,187.50	$ 2,275.00
20	2025	October	A	1,300	$ 60.00	36.00	17.50	3.50	$ 78,000.00	$ 46,800.00	$ 22,750.00	$ 4,550.00
21	2025	October	B	600	$ 100.00	80.00	18.75	3.50	$ 60,000.00	$ 48,000.00	$ 11,250.00	$ 2,100.00
22	2025	November	A	1,380	$ 60.00	36.00	17.50	3.50	$ 82,800.00	$ 49,680.00	$ 24,150.00	$ 4,830.00
23	2025	November	B	550	$ 100.00	80.00	18.75	3.50	$ 55,000.00	$ 44,000.00	$ 10,312.50	$ 1,925.00
24	2025	December	A	1,560	$ 60.00	36.00	17.50	3.50	$ 93,600.00	$ 56,160.00	$ 27,300.00	$ 5,460.00
25	2025	December	B	500	$ 100.00	80.00	18.75	3.50	$ 50,000.00	$ 40,000.00	$ 9,375.00	$ 1,750.00

Sheet1

STEP 2: There are different ways to create comparative contribution margin income statements for 2025. Here is one option:

- Go to the **Data** tab and select **Sort**. A pop-up window will appear. Verify the default **My data has headers** has been checked. Select **Product** in the first drop down list to ensure the data set is sorted properly.
- Insert two rows between Product A and Product B data to help separate this data (**Illustration 10.27**).

ILLUSTRATION 10.27 Results of Step 2

	A	B	C	D	E	F	G	H	I	J	K	L
1	Year	Month	Product	SumUnits Sold	Price	VCunit	Traceable FCunit	Common FCunit	Total Revenues	Total Direct VC	Total Applied TraceableFC	TotalApplied CommonFC
2	2025	January	A	600	$ 60.00	36.00	17.50	3.50	$ 36,000.00	$ 21,600.00	$ 10,500.00	$ 2,100.00
3	2025	February	A	700	$ 60.00	36.00	17.50	3.50	$ 42,000.00	$ 25,200.00	$ 12,250.00	$ 2,450.00
4	2025	March	A	760	$ 60.00	36.00	17.50	3.50	$ 45,600.00	$ 27,360.00	$ 13,300.00	$ 2,660.00
5	2025	April	A	740	$ 60.00	36.00	17.50	3.50	$ 44,400.00	$ 26,640.00	$ 12,950.00	$ 2,590.00
6	2025	May	A	820	$ 60.00	36.00	17.50	3.50	$ 49,200.00	$ 29,520.00	$ 14,350.00	$ 2,870.00
7	2025	June	A	900	$ 60.00	36.00	17.50	3.50	$ 54,000.00	$ 32,400.00	$ 15,750.00	$ 3,150.00
8	2025	July	A	980	$ 60.00	36.00	17.50	3.50	$ 58,800.00	$ 35,280.00	$ 17,150.00	$ 3,430.00
9	2025	August	A	1,040	$ 60.00	36.00	17.50	3.50	$ 62,400.00	$ 37,440.00	$ 18,200.00	$ 3,640.00
10	2025	September	A	1,220	$ 60.00	36.00	17.50	3.50	$ 73,200.00	$ 43,920.00	$ 21,350.00	$ 4,270.00
11	2025	October	A	1,300	$ 60.00	36.00	17.50	3.50	$ 78,000.00	$ 46,800.00	$ 22,750.00	$ 4,550.00
12	2025	November	A	1,380	$ 60.00	36.00	17.50	3.50	$ 82,800.00	$ 49,680.00	$ 24,150.00	$ 4,830.00
13	2025	December	A	1,560	$ 60.00	36.00	17.50	3.50	$ 93,600.00	$ 56,160.00	$ 27,300.00	$ 5,460.00
14												
15												
16	2025	January	B	1,710	$ 100.00	80.00	18.75	3.50	$ 171,000.00	$ 136,800.00	$ 32,062.50	$ 5,985.00
17	2025	February	B	1,640	$ 100.00	80.00	18.75	3.50	$ 164,000.00	$ 131,200.00	$ 30,750.00	$ 5,740.00
18	2025	March	B	1,360	$ 100.00	80.00	18.75	3.50	$ 136,000.00	$ 108,800.00	$ 25,500.00	$ 4,760.00
19	2025	April	B	1,270	$ 100.00	80.00	18.75	3.50	$ 127,000.00	$ 101,600.00	$ 23,812.50	$ 4,445.00
20	2025	May	B	1,200	$ 100.00	80.00	18.75	3.50	$ 120,000.00	$ 96,000.00	$ 22,500.00	$ 4,200.00
21	2025	June	B	940	$ 100.00	80.00	18.75	3.50	$ 94,000.00	$ 75,200.00	$ 17,625.00	$ 3,290.00
22	2025	July	B	860	$ 100.00	80.00	18.75	3.50	$ 86,000.00	$ 68,800.00	$ 16,125.00	$ 3,010.00
23	2025	August	B	720	$ 100.00	80.00	18.75	3.50	$ 72,000.00	$ 57,600.00	$ 13,500.00	$ 2,520.00
24	2025	September	B	650	$ 100.00	80.00	18.75	3.50	$ 65,000.00	$ 52,000.00	$ 12,187.50	$ 2,275.00
25	2025	October	B	600	$ 100.00	80.00	18.75	3.50	$ 60,000.00	$ 48,000.00	$ 11,250.00	$ 2,100.00
26	2025	November	B	550	$ 100.00	80.00	18.75	3.50	$ 55,000.00	$ 44,000.00	$ 10,312.50	$ 1,925.00
27	2025	December	B	500	$ 100.00	80.00	18.75	3.50	$ 50,000.00	$ 40,000.00	$ 9,375.00	$ 1,750.00

2

Sheet1

STEP 3: Enter a sum formula in cells D14 and D28 for total units sold for each product. The formula in the D14 cell is = sum(D2:D13). The other formulas follow the same pattern with their corresponding column letter name. The meaning of the calculated field in D14 = total Product A units sold and D28 = total Product B units sold. Then, in rows 14 and 28 enter similar sums for Columns I, J, K, and L (**Illustration 10.28**).

ILLUSTRATION 10.28 Results of Step 3

	A	B	C	D	E	F	G	H	I	J	K	L
1	**Year**	**Month**	**Product**	**SumUnits Sold**	**Price**	**VCunit**	**Traceable FCunit**	**Common FCunit**	**Total Revenues**	**Total DirectVC**	**TotalApplied TraceableFC**	**TotalApplied CommonFC**
2	2025	January	A	600	$ 60.00	36.00	17.50	3.50	$ 36,000.00	$ 21,600.00	$ 10,500.00	$ 2,100.00
3	2025	February	A	700	$ 60.00	36.00	17.50	3.50	$ 42,000.00	$ 25,200.00	$ 12,250.00	$ 2,450.00
4	2025	March	A	760	$ 60.00	36.00	17.50	3.50	$ 45,600.00	$ 27,360.00	$ 13,300.00	$ 2,660.00
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6	2025	May	A	820	$ 60.00	36.00	17.50	3.50	$ 49,200.00	$ 29,520.00	$ 14,350.00	$ 2,870.00
7	2025	June	A	900	$ 60.00	36.00	17.50	3.50	$ 54,000.00	$ 32,400.00	$ 15,750.00	$ 3,150.00
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10	2025	September	A	1,220	$ 60.00	36.00	17.50	3.50	$ 73,200.00	$ 43,920.00	$ 21,350.00	$ 4,270.00
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12	2025	November	A	1,380	$ 60.00	36.00	17.50	3.50	$ 82,800.00	$ 49,680.00	$ 24,150.00	$ 4,830.00
13	2025	December	A	1,560	$ 60.00	36.00	17.50	3.50	$ 93,600.00	$ 56,160.00	$ 27,300.00	$ 5,460.00
14				12,000					$ 720,000.00	$ 432,000.00	$ 210,000.00	$ 42,000.00
15												
16	2025	January	B	1,710	$ 100.00	80.00	18.75	3.50	$ 171,000.00	$ 136,800.00	$ 32,062.50	$ 5,985.00
17	2025	February	B	1,640	$ 100.00	80.00	18.75	3.50	$ 164,000.00	$ 131,200.00	$ 30,750.00	$ 5,740.00
18	2025	March	B	1,360	$ 100.00	80.00	18.75	3.50	$ 136,000.00	$ 108,800.00	$ 25,500.00	$ 4,760.00
19	2025	April	B	1,270	$ 100.00	80.00	18.75	3.50	$ 127,000.00	$ 101,600.00	$ 23,812.50	$ 4,445.00
20	2025	May	B	1,200	$ 100.00	80.00	18.75	3.50	$ 120,000.00	$ 96,000.00	$ 22,500.00	$ 4,200.00
21	2025	June	B	940	$ 100.00	80.00	18.75	3.50	$ 94,000.00	$ 75,200.00	$ 17,625.00	$ 3,290.00
22	2025	July	B	860	$ 100.00	80.00	18.75	3.50	$ 86,000.00	$ 68,800.00	$ 16,125.00	$ 3,010.00
23	2025	August	B	720	$ 100.00	80.00	18.75	3.50	$ 72,000.00	$ 57,600.00	$ 13,500.00	$ 2,520.00
24	2025	September	B	650	$ 100.00	80.00	18.75	3.50	$ 65,000.00	$ 52,000.00	$ 12,187.50	$ 2,275.00
25	2025	October	B	600	$ 100.00	80.00	18.75	3.50	$ 60,000.00	$ 48,000.00	$ 11,250.00	$ 2,100.00
26	2025	November	B	550	$ 100.00	80.00	18.75	3.50	$ 55,000.00	$ 44,000.00	$ 10,312.50	$ 1,925.00
27	2025	December	B	500	$ 100.00	80.00	18.75	3.50	$ 50,000.00	$ 40,000.00	$ 9,375.00	$ 1,750.00
28				12,000					$ 1,200,000.00	$ 960,000.00	$ 225,000.00	$ 42,000.00

Sheet1

STEP 4: Starting in Column N row 3, enter the row descriptions and column titles from Illustration 10.6. Then, enter the cell address and formulas in each cell. Please note that column R has been removed. Adjust the column widths and properly format the data cells (**Illustration 10.29**).

ILLUSTRATION 10.29 Formatting the Table for Illustration 10.6 (With Cell Formulas)

	N	O	P	Q	S
4		**2025 Income Statement**			
5		**ProductA**	**ProductB**	**Both Product Lines**	**Only ProductA**
6	Sales Revenues	+I14	+I28	=O6+P6	+O6
7	Less: Variable Costs	+J14	+J28	=O7+P7	+O7
8	Contribution Margin	=O6–O7	=P6–P7	=Q6–Q7	+O8
9	Less: Avoidable Fixed				
10	Expenses	+K14	+K28	+O10+P10	+O10
11	Segment Margin	=O8–O10	=P8–P10	=Q8–Q10	+O11
12	Less: Unavoidable Common				
13	Fixed Expenses	+L14	+L28	=O13+P13	=O13+P13
14	Net Income	**=O11–O13**	**=P11–P13**	**=Q11–Q13**	**=S11–S13**

Sheet1

The completed Excel table should look equivalent to Illustration 10.6.

STEP 5: Self-reflection: Consider how the analysis could contribute to the decision to keep or drop Product B. Are there any limitations to the analysis that should be considered?

- The results indicate this data has data veracity for the objective of deciding whether to drop Product B. By creating the comparative income statements in Illustration 10.6, we can analyze the contribution margin and segment margin, in addition to the net income generated by each product.
- Since many people are more proficient in Excel compared to other analysis tools, we may conclude that Excel was the best tool for efficiently building the income statement information models. We were able to calculate all totals quickly with cell references or easy formulas, and it is easy to format the table with both the text labels and the currency formatting in the table cells.
- It might be difficult for another employee or auditor to verify these calculations. They would have to examine each formula cell to determine if the calculations were appropriate or accurate. Since we cannot know what another person was thinking and deciding when creating this kind of information model spreadsheet, the finished spreadsheet does not provide the best documentation for someone else to understand, verify, or learn from. Additionally, the work we did for this spreadsheet could not be re-used with a new data set.

How To

HOW TO 10.2 Use Process Automation to Analyze Product Contributions to Net Income

How To 10.1 demonstrated how to use Excel to create the analysis shown in Illustration 10.6. What if you wanted to create an automated process to generate the same illustration? An automated data analysis tool, Alteryx, can be used to create a comparative income statement.

What You Need: Data The How To 10.2 data file. (Ask your instructor how to access the Alteryx software and how to name the results files.)

STEP 1: Open Alteryx and select the **Favorites** tab at the top of the screen. This set of tool icons contains the tools necessary for recreating Illustration 10.6.

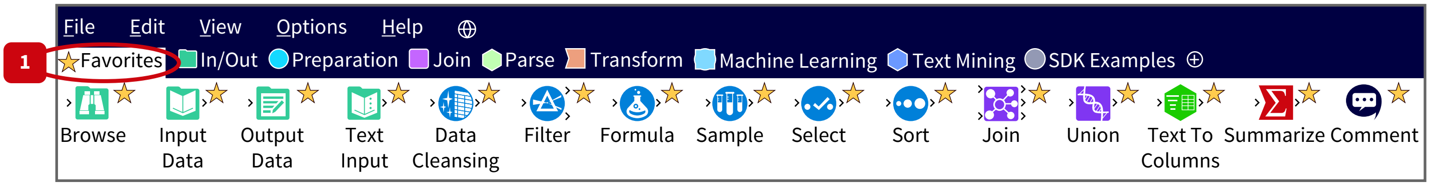
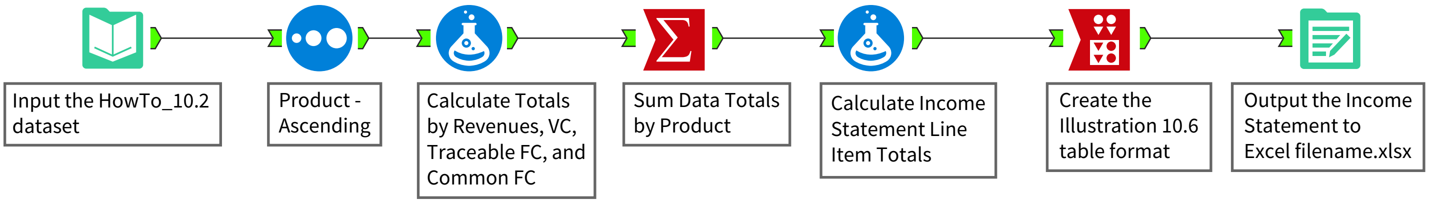


Illustration 10.30 is the completed, correct final Alteryx workflow. Use it as a guide as you work through the steps of this analysis.

ILLUSTRATION 10.30 Alteryx Workflow for Creating Comparative Income Statements



STEP 2: Add the **Input Data** tool to the workflow by dragging the icon into the biggest window on the Alteryx screen:

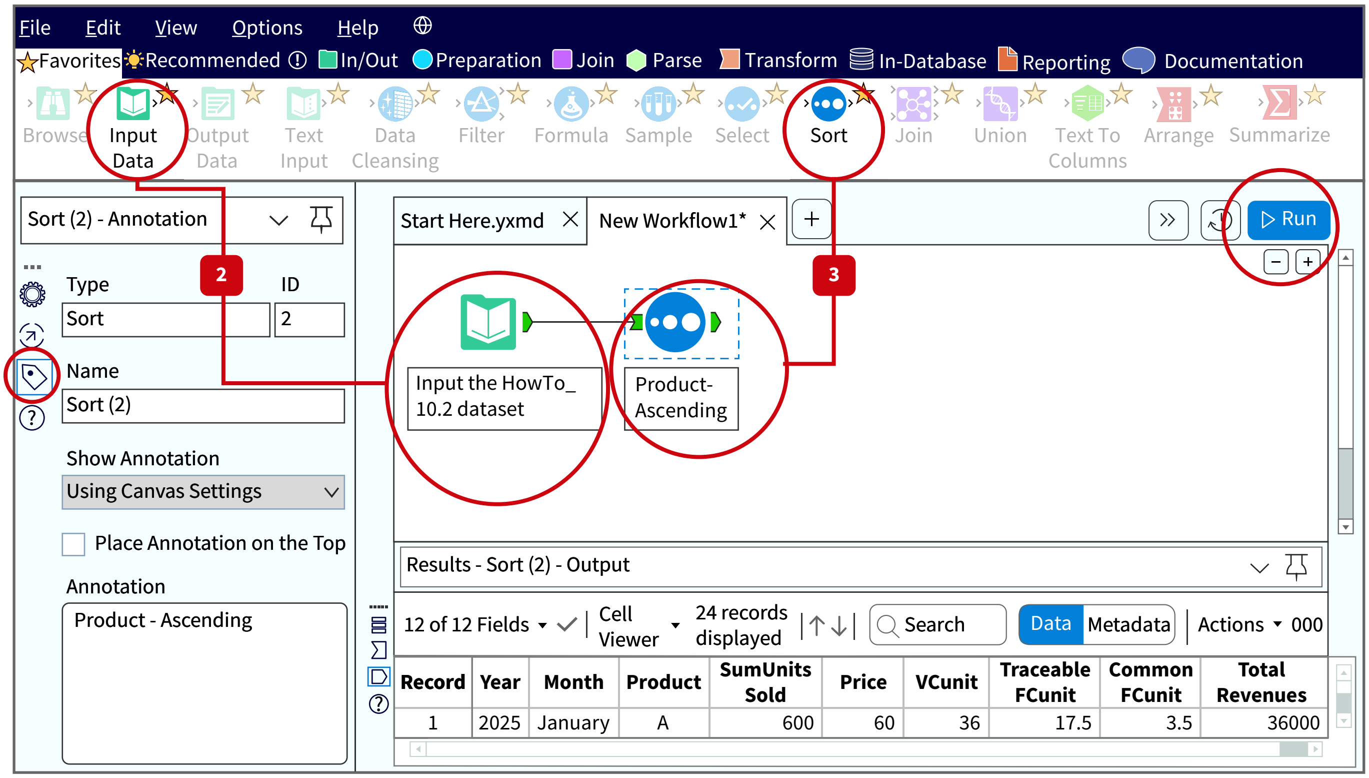
- Then, in the box under **Connect a File or Database**, select the down arrow. Import Sheet1 of the Excel data set.
- Run the input tool by clicking the blue **Run** button at the top right.

STEP 3: Add the **Sort** tool by dragging the icon into the workflow window to the right of the input icon.

- In the top left window, sort on the **Product** column in ascending order.
- Add the annotation (a comment of your choice) by clicking on the small luggage tag icon, which is the third icon on the left of the screen.
- Run the sort (**Illustration 10.31**).

Notice the **Results** window at the bottom right. The second icon on the far left will allow you to look at the data being input into the Sort step. The third icon shows the data results after the Sort function has run. In this view, the Product column rows have been sorted, and the Product A rows are above the Product B rows. It is useful to remember these two icons, as they can be key for problem-solving.

ILLUSTRATION 10.31 Alteryx Workflow after Step 3



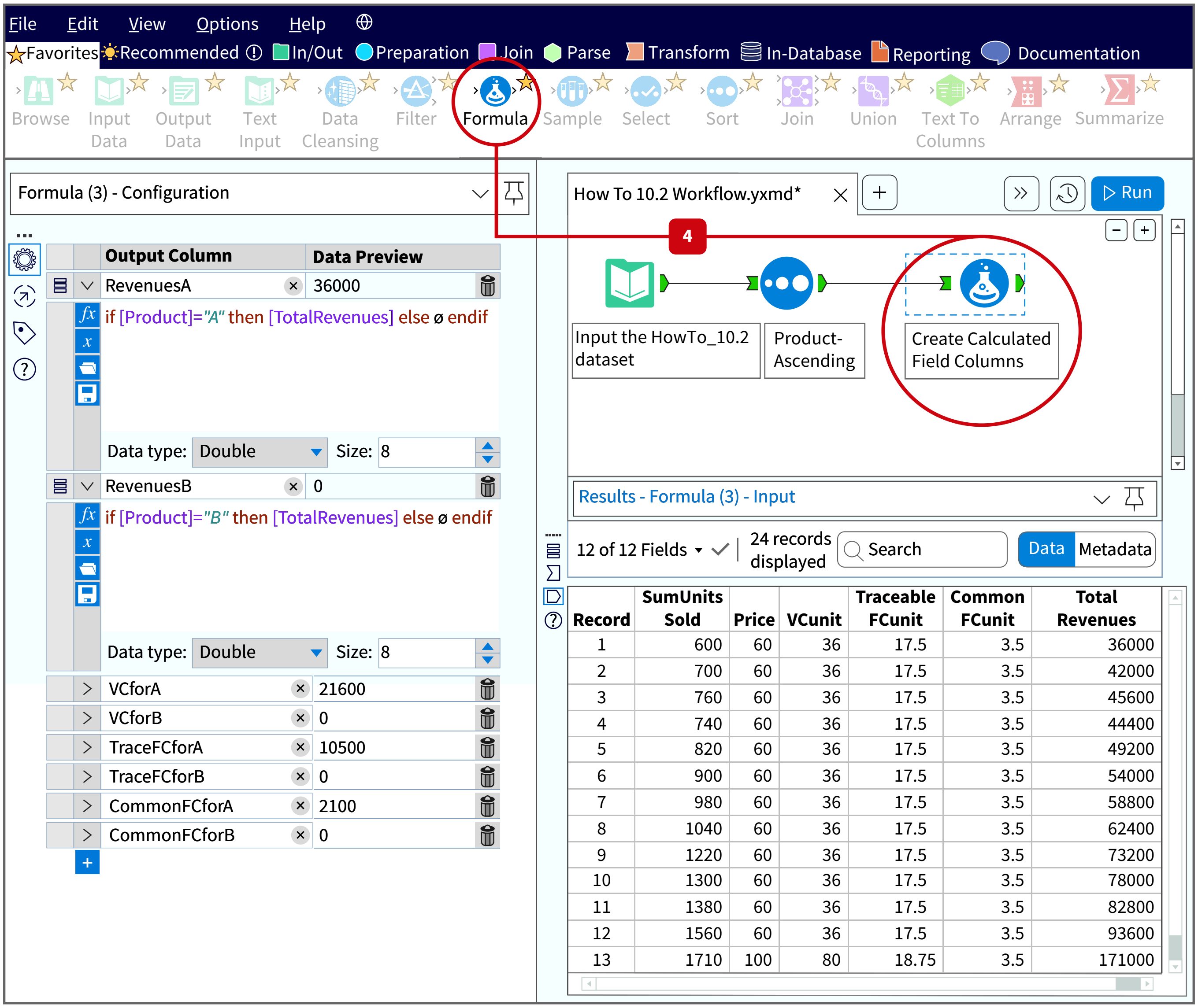
STEP 4: Attach the **Formula** icon to the workflow. Make the following new columns, and type in the formulas for each new column in the top left window:

- Start by selecting the arrow on the **Select Column** tab under the **Output Column** header.
- Choose **Add a new column**. Then, type each new column name in that same field.
- Below that, type the formula in the expression space, and select the proper data type (for all the column fields you build in this exercise, select the type **Double**).

New Column	Formula for the New Column
RevenuesA	if [Product]="A" then [TotalRevenues] else 0 endif
RevenuesB	if [Product]="B" then [TotalRevenues] else 0 endif
VCforA	if [Product]="A" then [TotalDirectVC] else 0 endif
VCforB	if [Product]="B" then [TotalDirectVC] else 0 endif
TraceFCforA	if [Product]="A" then [TotalAppliedTraceableFC] else 0 endif
TraceFCforB	if [Product]="B" then [TotalAppliedTraceableFC] else 0 endif
CommonFCforA	if [Product]="A" then [TotalAppliedCommonFC] else 0 endif
CommonFCforB	if [Product]="B" then [TotalAppliedCommonFC] else 0 endif

Add the annotation of your choice (small luggage tag icon) and run the formula tool (**Illustration 10.32**).

ILLUSTRATION 10.32 Alteryx Workflow After Step 4.



STEP 5: Attach the **Summarize** icon to the right end of the workflow:

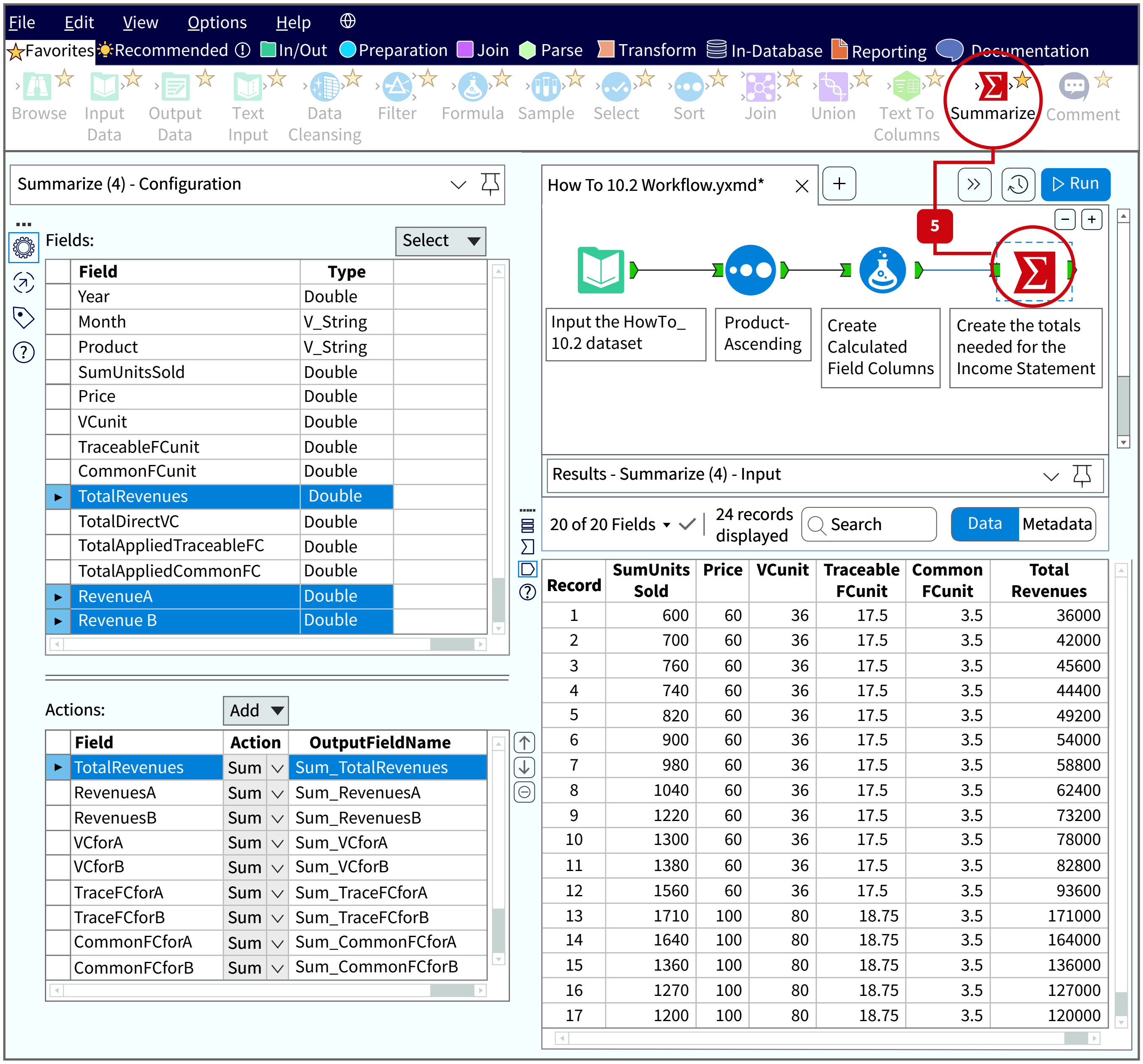
- In the top left window, select the fields listed in the left column of the table (you may have to scroll down to see them). Select a field by moving the cursor to the blank field to the left of the field name, hold down the control key, and left click on each field you want to select (you should select nine fields).

- Click **Add** and select **SUM**. These fields will automatically populate in the lower section, so you will not have to type the names (shown in the right column of the following table) in the Output field.

Field	Action	Output Field Name
TotalRevenues	Sum	Sum_TotalRevenues
RevenuesA	Sum	Sum_RevenuesA
RevenuesB	Sum	Sum_RevenuesB
VCforA	Sum	Sum_VCforA
VCforB	Sum	Sum_VCforB
TraceFCforA	Sum	Sum_TraceFCforA
TraceFCforB	Sum	Sum_TraceFCforB
CommonFCforA	Sum	Sum_CommonFCforA
CommonFCforB	Sum	Sum_CommonFCforB

Add the annotation of your choice (small luggage tag) and run the **Summarize** tool (**Illustration 10.33**).

ILLUSTRATION 10.33 Alteryx Workflow After Step 5



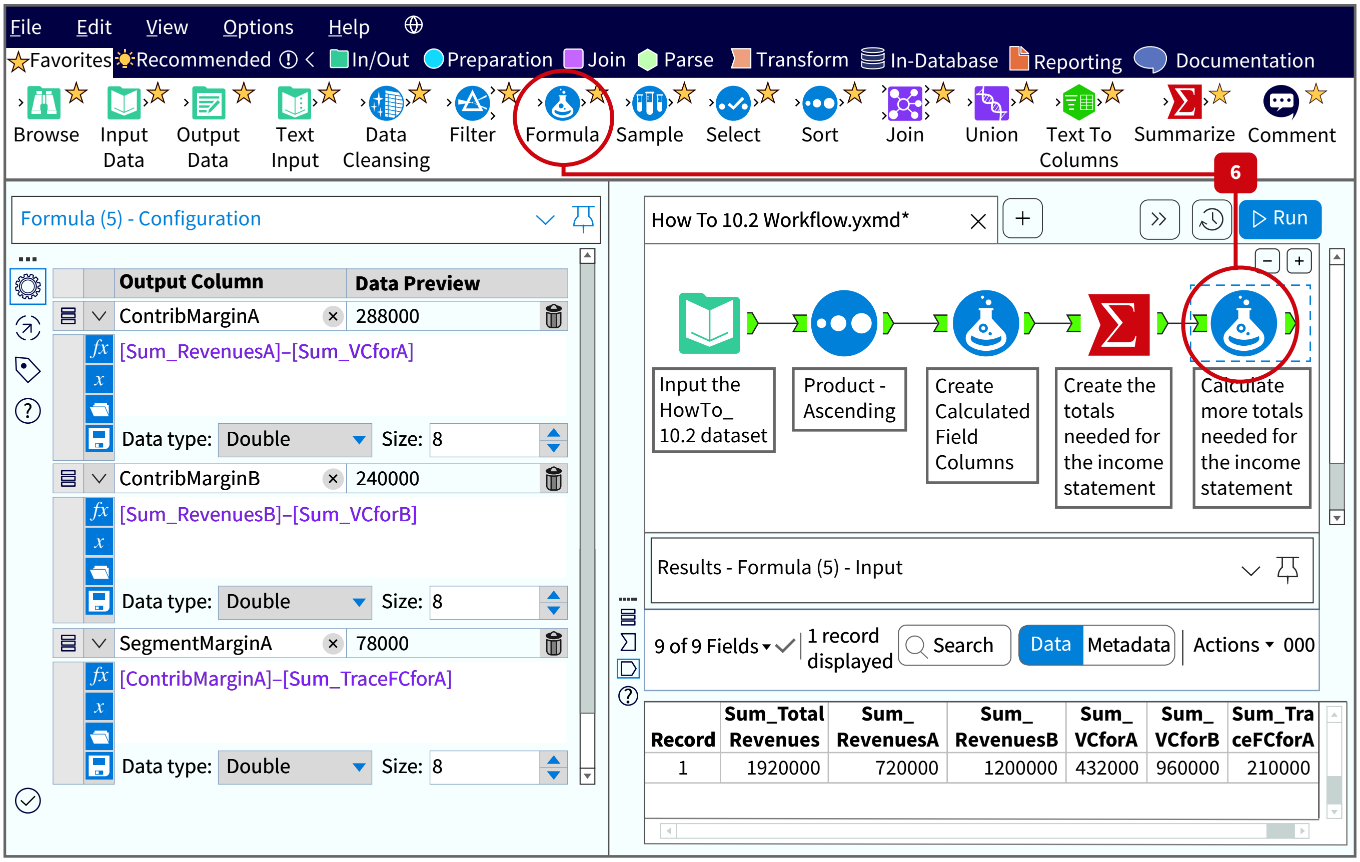
STEP 6: Attach the **Formula** icon to the right of the workflow:

- Make the following 13 new columns with their corresponding formulas. (Hint: Save time by selecting the field name from the dropdown list that appears when you type the bracket. This helps avoid typos, which could cause an error when you run the Formula function.) All new column types should be changed to **Double**.

New Column	Formula for the New Column
ContribMarginA	[Sum_RevenuesA]-[Sum_VCforA]
ContribMarginB	[Sum_RevenuesB]-[Sum_VCforB]
SegmentMarginA	[ContribMarginA]-[Sum_TraceFCforA]
SegmentMarginB	[ContribMarginB]-[Sum_TraceFCforB]
NetIncomeA	[SegmentMarginA]-[Sum_CommonFCforA]
NetIncomeB	[SegmentMarginB]-[Sum_CommonFCforB]
TotalVC	[Sum_VCforA]+[Sum_VCforB]
TotalContribMargin	[ContribMarginA]+[ContribMarginB]
TotalTraceFC	[Sum_TraceFCforA]+[Sum_TraceFCforB]
TotalSegmentMargin	[SegmentMarginA]+[SegmentMarginB]
TotalCommonFC	[SumCommonFCforA]+[Sum_CommonFCforB]
TotalNetIncome	[NetIncomeA]+[NetIncomeB]
NetIncomeOnlyA	[SegmentMarginA]-[TotalCommonFC]

- Add the annotation of your choice and run the formula tool (**Illustration 10.34**).

ILLUSTRATION 10.34 Alteryx Workflow After Step 6 (showing only the first three formulas)



STEP 7: Attach the **Arrange** icon to the workflow (it can be found under the **Transform** menu, not the **Favorites** menu):

- In the **Key Fields** window at the top left, make sure all the key fields have been de-selected, as those fields should not be columns in the income statement table (they should be de-selected already, but ensuring they are de-selected is a control).
- Below the section with these key fields, click the arrow to the right of **Columns**, and select **Add**. A new window will appear in the middle of the screen.

In this step, you will be adding the four income statement columns and the seven income statement rows (they are called Output fields in Alteryx):

	Product A	Product B	Both Products	Only Product A
Sales Revenues				
Variable Costs				
Contribution Margin				
Traceable Fixed Costs				
Segment Margin				
Common Fixed Costs				
Net Income				

Illustration 10.35 contains the information that should be entered for each income statement column.

ILLUSTRATION 10.35 Input Fields for Alteryx Arrange Step

Output Column Header	Description Field	Fields to Select
Product A	Add New Description	SumRevenuesA
		SumVCforA
		ContribMarginA
		SumTraceFCforA
		SegmentMarginA
		SumCommonFCforA
		NetIncomeA
Product B	None	SumRevenuesB
		SumVCforB
		ContribMarginB
		SumTraceFCforB
		SegmentMarginB
		SumCommonFCforB
		NetIncomeB
Both Products	None	SumTotalRevenues
		TotalVC
		TotalContribMargin
		TotalTraceFC
		TotalSegmentMargin
		TotalCommonFC
		TotalNetIncome

(Continued)

ILLUSTRATION 10.35 *(Continued)*

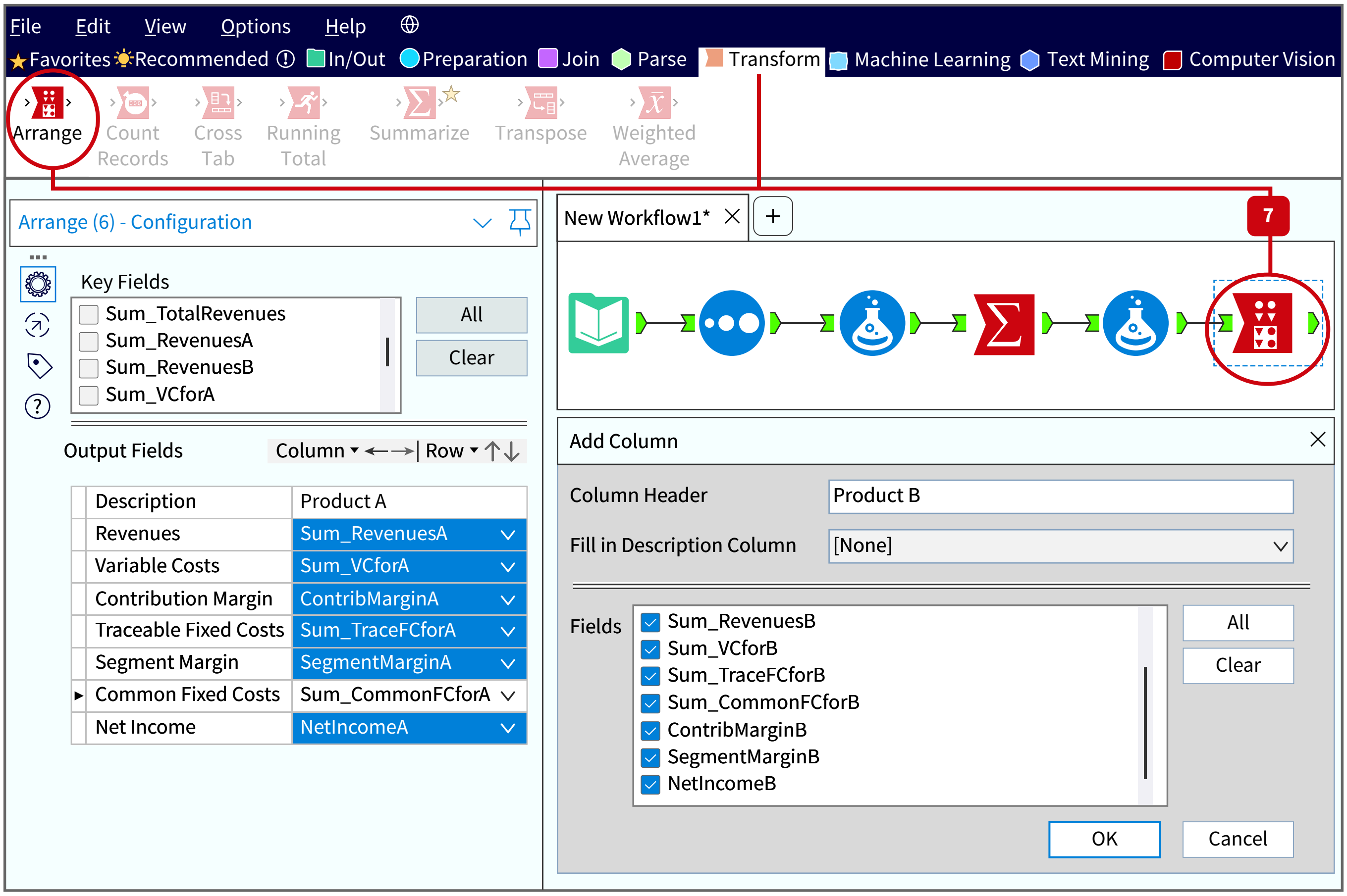
Output Column Header	Description Field	Fields to Select
Only Product A	None	SumRevenuesA
		SumVCforA
		ContribMarginA
		SumTraceFCforA
		SegmentMarginA
		TotalCommonFC
		NetIncomeOnlyA

- In the new window's **Column Header** field name, enter Product A. In the next field, select **Add New Description**. Note that the **Description** column only needs to be filled out for this first Product A column. For the other columns, select the Description option **None**.
- In the **Fields** section, select the fields indicated for this column and subsequently, for each of the other columns needed for the income statement. Repeat this process by adding columns for Product B, Both Products, and Only Product A columns.

In the left window, find the **Output Fields** section, where the **Description** column has a blue highlight. In the Product A column to its left, examine the vertical order of the fields, and re-arrange them in the correct vertical income statement order:

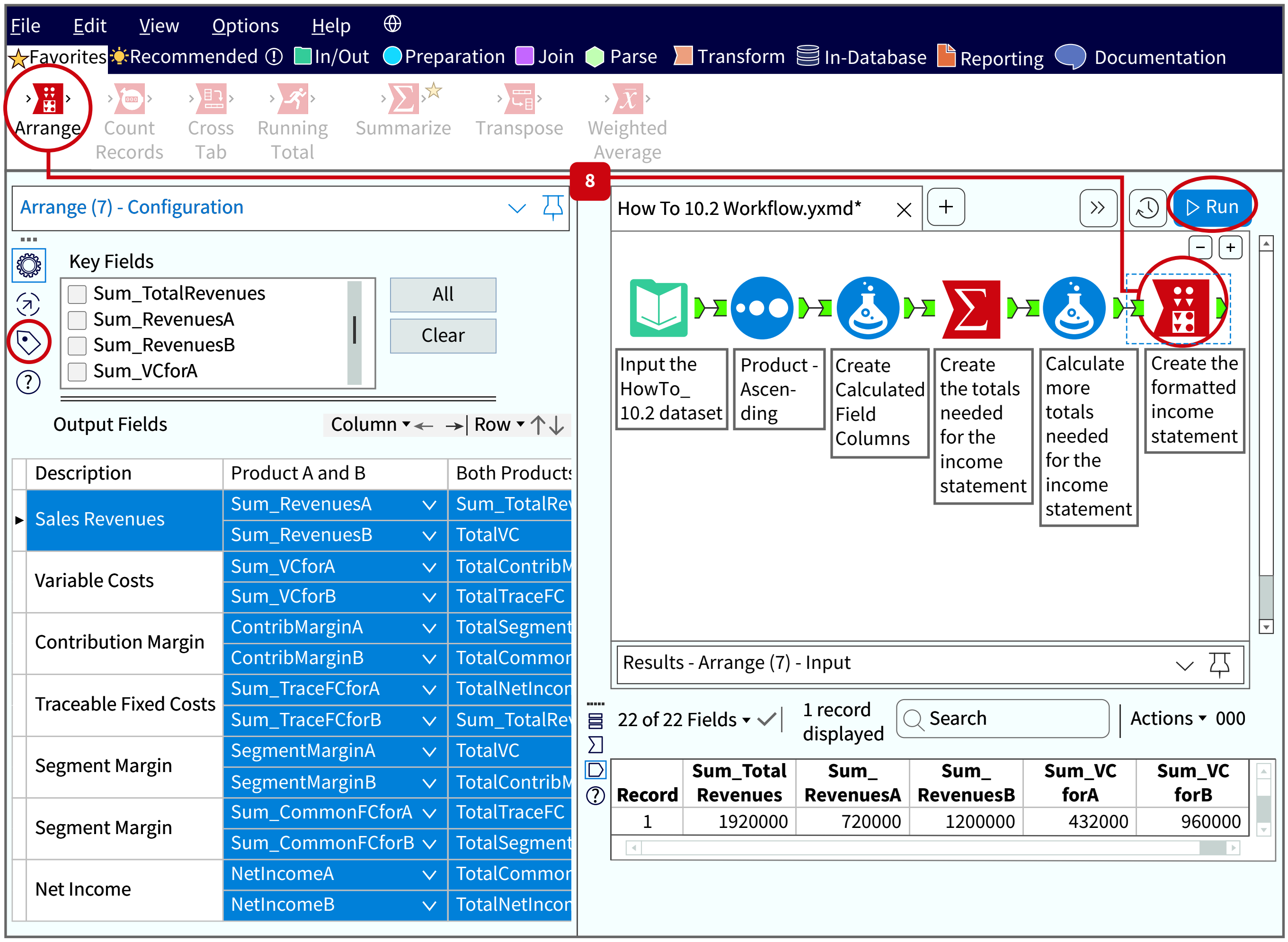
- Select the pull-down menu to the right of the field name and select the field that corresponds to the correct row order of the income statement.
- The correct order is listed in the left section of Illustration 10.36 as a guide.

ILLUSTRATION 10.36 Alteryx Workflow Arrange Step 7 When Entering Product B Column.



STEP 8: In the resulting table, use the arrows to move the fields in each column up and down as needed so they are in the proper income statement order from top (Sales Revenues) to bottom (Net Income). Annotate as you please (the luggage tag icon), and run the **Arrange** tool (**Illustration 10.37**)

ILLUSTRATION 10.37 Alteryx Workflow After Step 8



STEP 9: Attach the **Output** icon to the workflow, which you will find in the **Favorites** menu:

- Select the pull-down menu to the right of **Write to File** or **Database** in the **Output Data** window on the left.
- Next, a window opens. At the bottom of the middle column, select **Microsoft Excel .xlsx** format. This will change the window so you can select where to save the output file and how to name it.
- When the **Select Excel Output** window appears, change the name of Sheet1 to "2025 Comparative Income Statement." Do not specify a range of cells. Compare your Excel file output to Illustration 10.6. Select the annotation of your choice, and run the **Output** tool. Your workflow should look like Illustration 10.6.

STEP 10: Self-reflection: Compare your Excel experience and solution (How To 10.1) to your Alteryx experience and solution (How To 10.2). You may feel that Alteryx took much longer to set up (and it may have been more frustrating if you were not familiar with Alteryx). You may note that Excel formatted the results as currency, while Alteryx did not. However, Alteryx provides valuable benefits over Excel:

- The Alteryx workflow documents and explains the data preparation and information modeling processes step by step. The steps and their order in Excel were not documented.
- Alteryx clearly names and defines the calculated data fields used as calculated field variables.
- Alteryx workflows are excellent documentation for auditors, and are effective training tools for employees. They also can be rerun with the same data or different data, such as the next period's inputs.
- New workflow icons can be easily added as branching off from any of the original workflow icons, and new icons can be added to the end of the existing workflow.
- Most accountants find it very easy to pick up, understand, and use another person's Alteryx workflows. This is not true of Excel.

Regardless of whether the comparative income statements were created with Excel or Alteryx, the resulting insights are valuable when deciding whether to drop a product line. We can conclude with confidence that this data have great data veracity and value for our objective.

Data The Data tag appears when the data required to answer a question or complete an exercise are available on Wiley's online learning platform.

Multiple Choice Questions

1. **(LO 1)** Anderson hypothesized there would be a direct relationship between his customers' sales volume and the positive tone in their product satisfaction survey comments. The analysis he would perform to compare the number of units sold and textual data would have high
   a. data variety.
   b. data volume.
   c. data veracity.
   d. data value.

2. **(LO 1)** Nicolas is a financial accountant for a global company that daily processes millions of purchases from thousands of vendors. As he prepares the accounts payable records for the external auditors, he is interested in the ending accounts payable's data
   a. veracity.
   b. value.
   c. variety.
   d. velocity.

3. **(LO 1)** XYM, Inc. wants to change from in-house servers to using cloud service providers to increase their data storage capacity. This is likely due to high
   a. data value.
   b. data veracity.
   c. data volume.
   d. data variety.

4. **(LO 1)** Elizabeth was asked to analyze how many Facebook stories refer to her company. The unstructured textual data she will be analyzing has high
   a. data volume.
   b. data variety.
   c. data velocity.
   d. data value.

5. **(LO 1)** Which of the following is *not* likely a resource for a professional to stay up-to-date regarding data privacy regulation?
   a. The European Union's General Data Protection Regulation (GDPR)
   b. United States' Health Information Privacy Protection Act and the Health Insurance Portability and Accountability Act (HIPAA)
   c. California's Consumer Privacy Act (CCPA)
   d. The Sarbanes-Oxley Act of 2002 (SOX)

6. **(LO 1)** When the results of data analysis do not provide significant business intelligence to an organization, the data is not considered to have
   a. data volume.
   b. data value.
   c. data variety.
   d. data veracity.

7. **(LO 2)** Which of the following presents the correct order of the data analyses value creation chain?
   a. Data analysis, new intelligence, strategy shifts, value creation.
   b. New intelligence, data analysis, strategy shifts, value creation
   c. Value creation, strategy shifts, data analysis, new intelligence
   d. Data analysis, data mining, data processing, data textual analysis, value creation

8. **(LO 2)** The analysis stage of the data analytics process is currently enhanced by
   a. greater variety of data and combinations of data types.
   b. new accounting regulations.
   c. financial statement totals.
   d. comparison to last year's results.

9. **(LO 2)** Data mining techniques are being used by the profession to
   a. determine the purpose of data analysis.
   b. prepare the data for data analysis.
   c. prepare the data and explore the data further.
   d. communicate the data analysis findings to stakeholders.

10. **(LO 2)** Cognitive technologies include each of the following *except* for

- **a.** artificial neural networks.
- **b.** data mining.
- **c.** machine learning.
- **d.** classification of accounts receivables' collectability.

11. **(LO 2)** Continuous auditing involves

- **a.** auditors being present at the client's office year-round.
- **b.** performing the same audit procedures each year end.
- **c.** embedded software modules that capture test transactions as they occur.
- **d.** None of these answer choices are correct.

12. **(LO 2)** Blockchain technologies are being heavily utilized for which of the following?

- **a.** Smart contracts helping revenue recognition and lease documentation.
- **b.** Estimation of bad debts.
- **c.** Accumulated depreciation schedules.
- **d.** Performance dashboard metrics.

13. **(LO 3)** AIS professionals are most likely responsible for performing data analyses to gain insight in which of the following?

- **a.** The value the accounting system provides to the organization's internal users and recipients of information outputs.
- **b.** The value the data in the system provides to creditors through the financial statements.
- **c.** The value the data in the system provides to investors through the financial reporting system.
- **d.** The value the data in the system provides to external auditors.

14. **(LO 3)** Auditors are embracing new data analyses options in all of the following areas *except*

- **a.** classifying unusual transactions using cognitive technologies.
- **b.** performing risk assessments with process mining technologies.
- **c.** testing transactions with data mining technologies.
- **d.** creating automatic performance reports for client management.

15 **(LO 3)** Which of the following summarizes why managerial accountants have quickly embraced new data and analyses options?

- **a.** They are subject to regulatory restrictions.
- **b.** They have a long, innovative tradition of using interdisciplinary data for problem solving and to identify opportunities.
- **c.** They are employees of the company.
- **d.** They evaluate trends in financial balances over time.

16. **(LO 3)** Tax accountants are using emerging analysis technologies for which of the following?

- **a.** Optimizing international transfer pricing choices.
- **b.** Determining applicable tax rates.
- **c.** Documenting property tax responsibilities.
- **d.** Outlining responsibilities in the engagement letter with their clients.

Review Questions

1. **(LO 1)** Describe data variety, and give two examples of structured data and unstructured data.

2. **(LO 1)** Discuss why professionals are increasingly including unstructured data in their analyses. Provide an example of data analysis incorporating unstructured data.

3. **(LO 1)** Describe data veracity and discuss why it is important for data analysis.

4. **(LO 1)** What is data velocity? Provide examples of sources that generate internal and external high velocity data.

5. **(LO 1)** Explain the challenges organizations face due to the increasing rate of growth in organizational data. How are organizations responding to these challenges?

6. **(LO 1)** Discuss the importance of analyzing data with high data value.

7. **(LO 1)** Identify and describe the three types of data clouds.

8. **(LO 2, 3)** Explain the data analyses value creation chain. Provide an example highlighting the four aspects of the value creation chain.

9. **(LO 2)** What is the goal of process mining? Provide an example of how accountants are using process mining.

10. **(LO 2)** Describe the benefits of continuous auditing processes.

11. **(LO 2)** What are some of the benefits of blockchain technologies? What is an example of blockchain use in accounting?

12. **(LO 3)** Discuss how cognitive technologies are used by tax professionals to generate insights.

13. **(LO 3)** How have new data and technology options motivated systems accountants to consider alternative ways to evaluate AIS performance?

14. **(LO 1, 3)** Discuss the importance of managerial accountants considering data veracity as they perform data analysis for decision-making.

15. **(LO 3)** Identify examples of how new data and analyses technologies are adding value to professionals in financial accounting.

Brief Exercises

BE 10.1 (LO 1) Accounting Information Systems You are a member of the AIS steering committee for your organization. The committee is deciding between several types of cloud data service options. What factors might help you choose the cloud data service that will best serve the needs of your organization?

BE 10.2 (LO 1) Tax Accounting Assume you are a senior tax accountant for U.S. Outdoor Adventures, a company with sales operations in 15 states. Your tax analyst has provided you with analysis results to support the company's annual sales tax compliance for each sales tax jurisdiction. You notice that the internal data used in the analysis only includes eight months of the year. With respect to data veracity, how can using eight months of data impact the conclusions reached in the analysis?

BE 10.3 (LO 1) For each statement, indicate which data characteristic it best describes: volume, variety, velocity, veracity, and value.

1. Susan, an audit staff member, performed an analysis whereby she combined comments from customers' reviews with warranty expenses for the period. This analysis is an example of data ______________.
2. Parker verified that he had a complete download of all accounts receivable activity data before he calculated the AR aging at the end of the period. Having complete data is an example of considering data ______________.
3. Jerome collected hourly temperature data outside his ice cream store to determine if there is a relationship between temperature and ice cream sales. Temperature data collected hourly each day is an example of data ______________.
4. Janel's company has decided to move from in-house servers to cloud-based servers due to the amount of data that the company is collecting and storing. Movement to cloud-based data storage will address the company's concerns related to data ______________.

BE 10.4 (LO 2) Match each statement to its related term. Each term can be used once, more than once, or not at all.

a. Data mining
b. Smart contracts
c. Robotic process automation
d. Process mining
e. Continuous auditing
f. Textual analysis

Statement	Term
1. When the client's weekly payroll journal entry value was outside the defined terms of normal authorized activity, the audit manager reviewed an automatically generated report.	
2. The financial accountant obtained an electronic file of the organization's press releases and related investor comments. The accountant analyzes the pessimistic and optimistic words to understand sentiment around the company's annual performance.	
3. The operations accounting manager reviewed an event log that documented how much time it takes to complete operational steps in sales, shipping, and sending invoices to customers.	
4. The tax accountant analyzed their client's online blog discussions to assess risk of tax evasion.	

BE 10.5 (LO 2) Managerial Accounting A managerial accountant analyzed production data to identify a product line that could realize cost savings if changes were made to its production strategies. The operations team implemented several changes to the product's production line, ultimately leading to costs savings for the manufacturing facility and product. Map these activities into the data analysis value chain for production efficiency. Link the following statements to the appropriate elements in the value chain.

a. Implement production modifications
b. Analysis of production data
c. Increase production efficiency
d. Identify production cost inefficiencies

BE 10.6 **(LO 2, 3)** Match each statement to the most appropriate professional practice area. Each practice area can be used once, more than once, or not at all.

a. Accounting information systems
b. Auditing
c. Financial accounting
d. Managerial accounting
e. Tax accounting

Statement	Practice Area
1. Steven engaged in data mining as part of the risk assessment process to determine if there were concerning patterns in his client's sales cycle data.	
2. Chi used process mining to examine systems delays and errors. Her analysis results help identify new policies to optimize the system functionality.	
3. Wynn used textual analysis tools to better classify his company's multi-jurisdictional transactions for compliance with new tax laws.	
4. Henry employs smart contracts for all new leases made by his company. The smart contracts will improve GAAP lease classifications and revenue recognition.	
5. Jamal automated the process of calculating depreciation expense for his company's fixed assets by creating an Excel macro. He runs this macro at the end of each month to create his adjusting entry to record depreciation expense.	
6. John used textual analysis software to identify sentiment in his client's press releases. His goal is to identify additional risks of material misstatements in his client's financial statements.	

BE 10.7 **(LO 2, 3)** Match each statement to the technology that the accounting professional will most likely choose to use. Each technology term can be used once, more than once, or not at all:

a. Data mining
b. Smart contracts
c. Robotic process automation
d. Process mining
e. Continuous auditing
f. Textual analysis
g. Cognitive technologies

Statement	Technology
1. A managerial accountant analyzes the comments made by customers when they return a product to identify where improvements can be made to return policy and procedures.	
2. An audit staff member examines sales returns data for the objective of discovering patterns among product type or delivery timing.	
3. To reduce processing time costs, the accounts payable (AP) manager wants the system to electronically match vendor invoices with their associated purchase orders and receiving reports instead of the current matching process performed by the AP clerk.	
4. An internal auditor examines patterns in employee reimbursements related to travel to identify unusual expenses.	

BE 10.8 **(LO 2, 3)** **Managerial Accounting** A restaurant chain's managerial accountant gathers data from customers' mobile device take-out orders. List several ethical implications associated with the restaurant using and selling this data.

Exercises

EX 10.1 **(LO 1)** **Accounting Information Systems** **Cloud-Based Data Storage** Assume you are an accounting information systems accountant at a healthcare company that prioritizes data ethics. Your company collects personally identifiable information about patients and providers, including data that is protected under The Health Insurance Portability and Accountability Act of 1996 (HIPAA). The company has grown recently due to acquisitions of physician practice groups, which increased the types and number of medical procedures performed. As a result, data volume has increased, and your company has decided that cloud-based storage will best serve the future data needs of the company.

Two vendors are finalists to provide that cloud-based storage. One vendor, PrivacyConcept, Inc. is offering a private cloud to host your company's data storage needs. A second vendor, SkyHighStorage, is offering a public cloud with appropriate internal controls to separate your company's data from its other customer's data. Write a memo to the CIO (Chief Information Officer) outlining the relevant data ethics considerations when using cloud services and for deciding between the two vendors.

EX 10.2 **(LO 1)** **Data** **Managerial Accounting** **Comparative Income Statement and Data Value** You are a managerial accountant for Water Sports, Inc., a manufacturer of inflatable and hard SUPS (stand up paddle boards). The VP of Operations is wondering whether Water Sports should drop an inflatable product line that has a net-loss performance. She has asked you to prepare a comparative income statement for two products–SUP Model SX (10'6" inflatable) and SUP Model FW (11' inflatable).

1. Create a comparative income statement for these two products.
2. Would eliminating one of the SUP models increase the company's profitability? Why or why not?
3. Consider the data value of the data set used in this analysis. What alternative analyses could you consider to address if you should drop a net-loss producing product?

EX 10.3 **(LO 1)** **Auditing** **Data Ethics** As an external auditor for a large public company client, your engagement team has implemented continuous audit modules into your client's AIS. These modules test and collect evidence of sales revenues transactions during the current year. Your engagement partner has asked you to draft a memo for the audit file to document the continuous auditing process steps, as well as the professional due care that will be taken, to keep the client's collected sales data confidential. An incomplete draft of the memo that is missing important components follows. Complete the memo when more information is needed.

March 15, 20XX

Memo for the Audit Files

RE: Continuous Audit of Revenue Data & Professional Due Care Considerations

This memo documents the engagement team's procedures to provide reasonable assurance regarding the client's revenue-cycle transactions and the engagement team's procedures to ensure that the client's data is kept confidential. First, we discuss the audit procedures and second, we discuss professional due care considerations.

Continuous Audit of Revenue Procedures

The client has provided us access to all sales transaction data in real-time by allowing the audit firm to connect its proprietary analytics software to the company's relational database. As revenue transactions are recorded in the client's database, the auditor's software automatically tests and captures the audit data.

We performed audit procedures on this data throughout the year, including the following descriptive statistics and diagnostic tests:

1. ____

Based on these descriptive and diagnostic tests, and comparing our results to past years, abnormal transactions or anomaly transactions were identified and further investigated by collecting and reviewing their related source documents, such as contracts, invoices, and customer payments.

Professional Due Care Considerations

Since sales and collection transactions have a high data volume of personally identifiable data, great care must be taken to ensure that the client's data is kept private.

For example, **2.** ____

EX 10.4 **(LO 1)** **Managerial Accounting** **Data Characteristics** You are the managerial accountant for a national chain fitness company that owns several gyms and health food stores in populated locations. Your company also offers nutrition and personal training services. You have been asked to analyze sales revenues and cost of goods sold by comparing this year to the prior year. Your data strategy is to analyze the following data fields and measures. For each, discuss its data characteristics in terms of data volume, variety, velocity, veracity, and value.

1. The quantity of products sold in the health food stores.
2. The number of memberships sold each month.
3. The percentage of members who book appointments with nutrition or personal training professionals.
4. Member evaluations about experiences with nutritional or personal training professionals, including star ratings and comments.

EX 10.5 **(LO 2)** **Accounting Information Systems** **Robotic Process Automation** Your company asked you to automate the process of performing the three-way match control in the expenditure cycle. Currently, the AP clerk manually matches the vendor's name, quantity, and price between the vendor invoice, the purchase order, and the receiving report. This manual process takes the AP clerk all day, leaving no time to perform other tasks. Discuss how RPA can automate this internal control to reduce the hours engaged in this task.

EX 10.6 **(LO 2)** **Data** **Managerial Accounting** **Textual Analysis** You work in the corporate office of Sassy Seasonings, a large restaurant chain. The operations team has implemented new kitchen management policies at its most popular location. To understand whether these policies have improved customer satisfaction over time, the manager asks you to analyze data collected as part of the customer satisfaction survey. This data contains the date of the review and the customers' food and service ratings.

1. Classify the customer's food rating key words into one of three groups: 1. Good, 2. Average, and 3. Poor.

   (Hint: First, sort the Customer Food Rating in alphabetical order by selecting all the data, not the headers, and checking the box for Headers in the Data. Then, create a new column for these food rating groups and manually code each row by typing one of the categories for each group: "1. Good", "2. Average", or "3. Poor" in the new column. Critical thinking tip: all coders have their own personal bias, so your coding may differ from others. A good control would be to have more than one coder and to analyze inter-coder reliability).
2. Create a frequency distribution table using the COUNTIF function in Excel.

   (Hint: Starting in cell G3, use the COUNTIF function to total each food rating into one of the three categories, such as =COUNTIF(D3:D91,"1. Good"). Create a total row to show the total number of customer food ratings by category.)
3. Next, create another frequency distribution table by month for each of the customer food rating categories. Create a pivot table on a new worksheet (name the worksheet "Frequency Table"), select the food rating groups as both rows and values, and then select Months as columns. Format the resulting table as necessary with column width changes.
4. Finally, create a line chart where months is on the x-axis and count of each rating is on the y-axis.

   (Hint: Create a pivot chart line graph using the frequency distribution table from question part 3. Select the pivot table rows where the Jan, Feb, and March data resides for columns A-E (Not the blue shaded header and Grand Total rows.) Then, create the line chart. You may need to switch the row/columns to present the months on the x-axis. Make any other formatting edits. Save the file.)
5. Based on your analyses, did the kitchen management policies implemented by your operations manager improve customer food ratings over these months?

EX 10.7 (LO 2) Data Financial Accounting **Process Automation** You are a financial accountant for Water Sports, Inc. Every month you create comparative income statements for your product lines in the U.S. GAAP format, a repeated process which consumes much of your time. You have decided to use process automation in Alteryx to generate comparative income statements faster and more efficiently.

1. Create a workflow in Alteryx to automate the process of creating comparative income statements with the following rows:
   - Sales Revenues
   - Less Cost of Goods Sold
   - = Gross Profit
   - Less Selling and Administrative Expenses
   - = Net Income
2. What is your interpretation of your analysis results? How would your interpretation be different if some of your production was not sold and still in ending inventory?

(Hint: Refer to How To 10.2 for the steps to create the workflow in Alteryx. Some of the Alteryx new columns and their formulas will be different from How To 10.2. For example, new columns include cost of goods sold rather than contribution margin, selling and administrative expenses rather than traceable fixed expenses, segment margin, and common fixed expenses. Assume that all variable costs and the traceable fixed costs are production costs that flow through inventory to cost of goods sold and that the common fixed costs are all selling and administrative expenses. Finally, assume that all products are sold in the period that they are produced and there are no ending inventory.)

EX 10.8 (LO 2) Data Auditing Financial Accounting **Robotic Process Automation** Your client, OneStopShop, Inc., is a public company with six independent business divisions: floor cleaning; lawn care; pest control; household cleaning; corporate office cleaning; and plumbing products. Your manager has asked you to calculate the following financial ratios for each of the divisions and compare their values between the current and previous years:

- Return on assets (net income / average total assets)
- Profit margin (net income / sales)
- Asset turnover (sales/ average total assets)
- Current ratio (current assets / current liabilities)

The client has provided a data set with select income statement and balance sheet data for each division. Because this is a task that you are likely to perform for other engagements, you decide to build a macro in Excel to calculate these financial statement ratios. Record a macro for the floor cleaning division. Then, run this macro on each of the divisions to calculate the ratios for the remaining divisions.

EX 10.9 (LO 2) Financial Accounting **Blockchain Technologies** You are a financial accountant in a large retail company whose supply chain partners want to use blockchain technologies to record all transactions. Explain the financial reporting benefits from using blockchain technology and factors that need to be considered for this decision.

EX 10.10 (LO 2, 3) Auditing Accounting Information Systems **Technology Developments** You are a systems accountant working with your company's internal auditor to analyze the compliance of your company's authorized purchase credit card (e.g., "P-card") transactions with company policies. A new analysis technology helps you easily identify the spending transactions by several employees that do not comply with the policies. Explain which technology development described in the chapter would best help you find and document where controls in the P-card process either must be improved or added to prevent these unauthorized transactions.

EX 10.11 (LO 2, 3) Financial Accounting **Technology Developments** You work for an investment manager who asks you to retrieve a large portfolio of different companies' closing stock market prices each trading day. Which data analysis technology would make this daily task more efficient?

EX 10.12 (LO 2, 3) Data Auditing Financial Accounting **Textual Analysis** As the external auditor for a computer manufacturing client, you are asked to examine the company's warranty expense and related accrual adjusting entry for GAAP compliance in valuation calculations. The client provided the following summary of warranty expense related data from the prior year (audited) and the current year (unaudited).

F/S	AcctNo	AcctDescription	6/30/2025	6/30/2024	Difference	%Change
Income Statement Account	32153	Warranty Expense	$ 347,954	$ 535,849	$ (187,895)	−35%
Balance Sheet Account	23829	Warranty Liability	$ 121,784	$ 241,132	$ (119,348)	−49%

Both the warranty expense and warranty liability decreased significantly compared to the prior year. You asked your client's financial accountant why there was such a drastic decrease in the expense and liability. The accountant responded, "We have had significant increases in product quality and therefore we don't have a lot of warranty claims these days. Customers are happier."

The objective was finding evidence to support or deny this explanation, so you obtained customer satisfaction text data about the computers from social media postings. You planned to perform textual analysis regarding whether there was an increase in positive customer sentiment about the computer product quality.

1. Classify the customer comments into three groups: 1 = low quality, 2 = neutral, and 3 = high quality.

   (Hint: Sort the product ratings or customer comments in alphabetical order to make it easier to assign categories to each comment. Use the COUNTIF function to total each comment into one of the three groups. Critical thinking note: Your coding may disagree with other coders, and you might be inconsistent within your own coding.)
2. Create a frequency distribution table that shows the total number of customer comments by group, by month, and in total for each of the customer comment groups.

   (Hint: Transform the date column into months from the Home menu. In the Number section, select Custom formatting, then type "mmm" in the type (no quotes). This does not change the date data, just its presentation, preserving your original dates. Then, create a pivot table using your groups as rows and values and months as columns.)
3. Create two time series visualizations to determine if customer sentiments about their computer products are improving or not. One visualization should use customer text comments and the other their product ratings.
4. Interpret these results and determine if any further audit steps are necessary to properly estimate the warranty expense and related accrual for the first six months. Suggest internal controls for the critical thinking risks mentioned in question 1.

Professional Application Case: Adventure Sports

Adventure Sports is a new chain of retail stores with both online and brick-and-mortar locations. They sell a variety of outdoor activity and sports products ranging from water sports to mountaineering and camping. The company is growing quickly, currently operating ten stores across five states. These stores are supplied with products from a distribution center that receives all its purchased products from vendors. Online sales are processed out of Adventure Sports' distribution center. The brick-and-mortar stores focus on high-quality customer service and offer classes for new outdoor sports enthusiasts. The company has developed strong relationships with its vendors and its customers. Customers tend to return to the stores, demonstrating significant customer loyalty.

Because Adventure Sports is growing so quickly, they want to automate more inventory purchasing processes to make them more efficient and effective, with better internal controls. What follows is an entity-relationship diagram (ERD) showing the tables and variables involved in the process.

VendorMasterFile	PurchaseOrder	ReceivingReport	Invoices
VendorID	PONo	RECNO	InvoiceNO
VendorName	VendorID	PONo	PONo
VendorAddress	VendorName	VendorID	VendorID
VendorCity	PODate	VendorName	VendorName
VendorState	POItemID	ItemID	ItemID
VendorZip	POItemDescription	ItemDescription	ItemDescription
VendorPayterms	POItemCost	RecItemQty	InvoiceQty
	POItemQty	RecDate	InvoiceItemCost
	POExpectedPreceiptDate		InvoiceTotal

PAC 10.1 Accounting Information Systems: Use Process Automation for Accounts Payable

Data **Accounting Information Systems** You are the systems accountant reviewing the accounts payable process to approve vendor invoices for payment. Currently, the process requires manual matching between the invoice, receiving report, and purchase order to approve the invoice for payment. You have interviewed the AP clerk and the financial accountant and noted that to approve an invoice for payment, the following conditions must be met:

- The invoice data must match the data included on the purchase order including, VendorID, ItemID, and Quantity.
- The invoice data must match the data included on the receiving report including VendorID, ItemID, and Quantity

Use Alteryx to build a workflow that results in two outputs: a file that is "Okay to Pay" invoices and a file that is "Exceptions."

PAC 10.2 Auditing: Continuous Auditing of Company Purchases

Auditing You are an audit senior on the Adventure Sports' annual audit. Your audit manager has asked you to develop a plan for the engagement team to use continuous audit methodologies to reduce the amount of audit work and testing performed at year-end. Your assigned audit area is the order-to-pay cycle. Draft a memo for the audit files discussing continuous auditing and outlining your plan for continuous monitoring of the purchasing data for your client.

PAC 10.3 Financial Accounting: Automate Procedures for Approving Vendor Payments

Data **Financial Accounting** **Accounting Information Systems** You are a financial accountant for Adventure Sports. Currently, your process for approving vendor invoices for payment is matching the hardcopy invoice from the vendor to the relevant purchase order and receiving data in the AIS:

- Purchase orders are keyed into the AIS in the PurchaseOrder table by the purchasing group after they are sent to the vendor.
- Receiving report data is keyed into the AIS in the receiving department after items are received by the vendor.
- Upon receipt of a vendor invoice, the invoice is keyed into the AIS VendorInvoices table and filed by due date in the hardcopy file in the AP clerk's office.
- Daily, the AP clerk pulls the invoices that are due within the next few days and performs the following:

  Step 1: For each invoice, the AP clerk looks up the corresponding PO number to the PO in the PurchaseOrders table. The AP clerk verifies that the quantity purchased was the quantity invoiced. If the quantities agree within a 2% tolerance level, then the AP clerk moves to the next step.

  Step 2: For each invoice, the AP clerk looks up the PO number that corresponds to the PO number in the ReceivingReport table. The AP clerk verifies that the quantity received was the quantity invoiced. If the quantities agree within a 2% tolerance level, then the clerk moves to the next step.

  Step 3: For each invoice that passes step 1 and 2, the AP clerk stamps "Okay to Pay" on the invoice and adds it to the Ok to Pay list, which is sent to the treasurer who prepares, signs, and sends checks.

1. Match and compare the quantities from the purchase orders, the receiving reports, and the vendor invoices. The matched invoice information within the 2% quantity deviation tolerance should be labeled as "Okay to Pay." Any invoices above the 2% quantity deviation should be labeled "Exception" so the AP clerk can follow up with the vendor.
2. Discuss the benefits of using Excel to increase efficiencies and potentially reduce errors in the approval of vendor invoices.

PAC 10.4 Managerial Accounting: Create a Vendor Performance Dashboard

Data **Managerial Accounting** **Accounting Information Systems** As a managerial accountant for Adventure Sports, you want to know if there are patterns in the purchasing data. Create a visualization dashboard to to measure the following metrics related to vendor purchases:

- The count of purchase orders for each vendor during this period.
- The percentage of purchase costs incurred during this period by vendor.
- The percentage of purchase costs incurred by product.

PAC 10.5 Tax Accounting: Explain Robotic Process Automation for Current Tax Rates

Tax Accounting Currently the data needed to complete the corporate tax return comes from several different data sources. Each month you download the data into Excel spreadsheets and use the Excel spreadsheet to complete the tax accrual analysis. Discuss how technology might be used to automate this process. Are there other technologies that Adventure Sports should consider for use in the tax department?

Le Grind Continuing Case: Prepare Gross Profit Data With an Alteryx Workflow

Data Visit Wiley's online learning platform for the case background, additional questions, data, and more information about the continuing case.